

Linux Commands Execution:

1. Create user with name Techie and provide sudo access to user.

```
poojith@Linux:/root$ su Techie
Password:
$ id
uid=1001(Techie) gid=1001(Techie) groups=1001(Techie)
$ su poojith
Password:
poojith@Linux:/root$ sudo su -
[sudo: authenticate] Password:
root@Linux:~# find / -name sudoers
/usr/share/doc/sudo/examples/sudoers
/usr/share/doc/sudo-common/sudoers
/snap/core24/1643/etc/sudoers
/snap/core24/1587/etc/sudoers
/etc/sudoers
find: '/run/user/1000/gvfs': Permission denied
find: '/run/user/1000/doc': Permission denied
root@Linux:~# vi /etc/sudoers
root@Linux:~# su Techie
$ sudo apt install git
[sudo: authenticate] Password:
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14 linux-tools-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  git

Installing dependencies:
  git-man liberror-perl

Suggested packages:
  git-doc git-email git-gui gitk gitweb git-cvs git-svn

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 90
  Download size: 5,453 kB
  Space needed: 28.4 MB / 22.5 GB available

Continue? [Y/n] S
```

2. Navigate to the home directory.

```
poojith@Linux:/root$ cd ~
poojith@Linux:~$ S
```

3. Create a new directory.
and
4. List the contents of a directory.

```
poojith@Linux:/root$ cd ~
poojith@Linux:~$ mkdir Ravutla
poojith@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Ravutla  Templates  Videos  snap
poojith@Linux:~$
```

5. Change the current directory.

and

6. Create a new empty file.

```
poojith@Linux:/root$ cd ~
poojith@Linux:~$ mkdir Ravutla
poojith@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Ravutla  Templates  Videos  snap
poojith@Linux:~$ cd Ravutla/
poojith@Linux:~/Ravutla$ touch hello
poojith@Linux:~/Ravutla$ ls
hello
poojith@Linux:~/Ravutla$ wc hello
0 0 0 hello
poojith@Linux:~/Ravutla$
```

7. View the contents of a file.

```
poojith@Linux:/root$ cd ~
poojith@Linux:~$ mkdir Ravutla
poojith@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Ravutla  Templates  Videos  snap
poojith@Linux:~$ cd Ravutla/
poojith@Linux:~/Ravutla$ touch hello
poojith@Linux:~/Ravutla$ ls
hello
poojith@Linux:~/Ravutla$ wc hello
0 0 0 hello
poojith@Linux:~/Ravutla$ vi hello
poojith@Linux:~/Ravutla$ cat hello
hi, hello good evening everyone.
poojith@Linux:~/Ravutla$
```

8. Copy a file to another location.

```
poojith@Linux:~$ pwd
/home/poojith
poojith@Linux:~$ cd ..
poojith@Linux:/home$ ls
poojith
poojith@Linux:/home$ cd
poojith@Linux:~$ ls
Desktop Documents Downloads Music Pictures Poojith Public Ravutla Templates Videos hello snap
poojith@Linux:~$ cp -r hello /home/poojith/Poojith
poojith@Linux:~$ ls
Desktop Documents Downloads Music Pictures Poojith Public Ravutla Templates Videos hello snap
poojith@Linux:~$ cd Poojith
poojith@Linux:~/Poojith$ ls
hello
poojith@Linux:~/Poojith$
```

9. Move a file to another location.

```
poojith@Linux:~/Poojith$ ls
hello
poojith@Linux:~/Poojith$ cd ..
poojith@Linux:~$ ls
Desktop Documents Downloads Music Pictures Poojith Public Ravutla Templates Videos hello snap
poojith@Linux:~$ pwd
/home/poojith
poojith@Linux:~$ mv hello Ravutla/
poojith@Linux:~$ cd Ravutla/
poojith@Linux:~/Ravutla$ ls
hello
poojith@Linux:~/Ravutla$ cd ..
poojith@Linux:~$ ls
Desktop Documents Downloads Music Pictures Poojith Public Ravutla Templates Videos snap
poojith@Linux:~$
```

10. Rename a file.

```
poojith@Linux:~/Poojith$ ls
hello
poojith@Linux:~/Poojith$ cd ..
poojith@Linux:~$ ls
Desktop Documents Downloads Music Pictures Poojith Public Ravutla Templates Videos hello snap
poojith@Linux:~$ pwd
/home/poojith
poojith@Linux:~$ mv hello Ravutla/
poojith@Linux:~$ cd Ravutla/
poojith@Linux:~/Ravutla$ ls
hello
poojith@Linux:~/Ravutla$ cd ..
poojith@Linux:~$ ls
Desktop Documents Downloads Music Pictures Poojith Public Ravutla Templates Videos snap
poojith@Linux:~$ cd Ravutla
poojith@Linux:~/Ravutla$ ls
hello
poojith@Linux:~/Ravutla$ mv hello hello_poojith
poojith@Linux:~/Ravutla$ ls
hello_poojith
poojith@Linux:~/Ravutla$
```

11. Delete a file.

```
poojith@Linux:~/Ravutla$ ls
hello_poojith
poojith@Linux:~/Ravutla$ touch hi
poojith@Linux:~/Ravutla$ ls
hello_poojith hi
poojith@Linux:~/Ravutla$ rm hi
poojith@Linux:~/Ravutla$ ls
hello_poojith
poojith@Linux:~/Ravutla$
```

12. Grant or revoke permissions on a file or directory.

```
poojith@Linux:~/Ravutla$ ls -l hello_poojith
-rw-rw-r-- 1 poojith poojith 33 Aug  7 10:38 hello_poojith
poojith@Linux:~/Ravutla$ chmod 776 hello_poojith
poojith@Linux:~/Ravutla$ ls -l hello_poojith
-rwxrwxrw- 1 poojith poojith 33 Aug  7 10:38 hello_poojith
poojith@Linux:~/Ravutla$ chmod 664 hello_poojith
poojith@Linux:~/Ravutla$ ls -l hello_poojith
-rw-rw-r-- 1 poojith poojith 33 Aug  7 10:38 hello_poojith
poojith@Linux:~/Ravutla$
```

13. View the current date and time.

```
poojith@Linux:~/Ravutla$ date
Fri Aug  7 11:27:39 AM UTC 2026
poojith@Linux:~/Ravutla$
```

14. Check the system uptime.

System uptime is the amount of time a computer or server has been **running continuously since its last boot (restart)**.

```
poojith@Linux:~/Ravutla$ uptime
11:31:07 up 2:08, 2 users, load average: 0.00, 0.02, 0.00
poojith@Linux:~/Ravutla$
```

15. View the running processes.

➔ To view the running processes we use the command **ps -ef**

1. **Ps** > process status
2. **-e** > every process
3. **-f** > full-format listing

Attaching the starting and ending part of the output as they are more.

```

Aug 8 10:04
root@kali:~/Desktop
ps -ef
USER          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0 09:30 ?        00:00:00 /usr/sbin/sshd -w -l -b -o -s -D -u root -p /etc/ssh/sshd_config
root           2        0  0 09:30 ?        00:00:00 [kthreadd]
root           3        2  0 09:30 ?        00:00:00 [kworker/0:0kpmq_releas]
root           4        2  0 09:30 ?        00:00:00 [kworker/0:0rcu_gp]
root           5        2  0 09:30 ?        00:00:00 [kworker/0:0rcu_perq_w]
root           6        2  0 09:30 ?        00:00:00 [kworker/0:0kfree_rcu_rela]
root           7        2  0 09:30 ?        00:00:00 [kworker/0:0slu]
root           8        2  0 09:30 ?        00:00:00 [kworker/0:0netio]
root          12        2  0 09:30 ?        00:00:00 [kworker/0:0irq-rcu_handler]
root          13        2  0 09:30 ?        00:00:00 [kworker/0:0rcu_gp_w]
root          14        2  0 09:30 ?        00:00:00 [ksftlq/0]
root          15        2  0 09:30 ?        00:00:00 [rcu_gp_wq]
root          16        2  0 09:30 ?        00:00:00 [rcu_exp_gp_w_kthread_worke]
root          17        2  0 09:30 ?        00:00:00 [migration/0]
root          18        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          19        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          20        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          21        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          22        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          23        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          24        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          25        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          26        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          27        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          28        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          29        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          30        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          31        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          32        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          33        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          34        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          35        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          36        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          37        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          38        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          39        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          40        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          41        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          42        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          43        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          44        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          45        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          46        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          47        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          48        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          49        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          50        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          51        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          52        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          53        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          54        2  0 09:30 ?        00:00:00 [kworker/0:0idle]
root          55        2  0 09:30 ?        00:00:00 [kworker/0:0idle]

Aug 8 10:05
root@kali:~/Desktop
ps -ef
USER          PID    PPID  C STIME TTY          TIME CMD
root          4242    3489  0 09:50 ?        00:00:00 /usr/libexec/gsd-printer
root          4247    4062  0 09:50 ?        00:00:00 /usr/libexec/ibus-dconf
root          4252    4062  0 09:50 ?        00:00:00 /usr/libexec/ibus-extension-gtk3
root          4268    3489  0 09:50 ?        00:00:00 /usr/libexec/ibus-portal
root          4292    3489  0 09:50 ?        00:00:00 /usr/bin/gjs -m /usr/share/gnome-shell/org.gnome.ScreenSaver
root          4302    3489  0 09:50 ?        00:00:00 /usr/libexec/gnome-daemon
root          4320    3489  0 09:50 ?        00:00:00 /usr/libexec/gnome-shell-volume-monitor
root          4330    3489  0 09:50 ?        00:00:00 /usr/libexec/gnome-identity-service
root          4341    3489  0 09:50 ?        00:00:00 /usr/libexec/localsearch
root          4342    3489  0 09:50 ?        00:00:00 /usr/bin/gnome-desktop-integration/30/usr/bin/nautilus-desktop-integration
root          4349    3489  0 09:50 ?        00:00:00 /usr/libexec/nautilus-desktop-portal
root          4360    3489  0 09:50 ?        00:00:00 /usr/libexec/evolution-alchemy-factory
root          4387    3489  0 09:50 ?        00:00:00 /usr/libexec/nautilus-desktop-gnome
root          4390    3489  0 09:50 ?        00:00:00 /usr/libexec/gvfs-goa-volume-monitor
root          4411    3489  0 09:50 ?        00:00:00 /usr/libexec/gvfs-sftp-volume-monitor
root          4427    3489  0 09:50 ?        00:00:00 /usr/libexec/gvfs-goa-volume-monitor
root          4450    3489  0 09:50 ?        00:00:00 /usr/libexec/gvfs-gphoto2-volume-monitor
root          4463    3489  0 09:50 ?        00:00:00 /usr/libexec/evolution-addressbook-factory
root          4471    4342  0 09:50 ?        00:00:00 /usr/bin/gnome-desktop-integration/30/usr/bin/nautilus-desktop-integration
root          4514        2  0 09:50 ?        00:00:00 [kworker/0:0ext-rx-conver]
root          4542        2  0 09:50 ?        00:00:00 [kworker/0:0ext-usb]
root          4557    4062  0 09:50 ?        00:00:00 /usr/libexec/ibus-engine-imple
root          4571        2  0 09:50 ?        00:00:00 [kworker/0:0ext-rx-con]
root          4577        2  0 09:50 ?        00:00:00 [kworker/0:0ext-rx-con]
root          4690    3489  0 09:50 ?        00:00:00 /usr/libexec/gvfsd-metadata
root          4700    3489  0 09:50 ?        00:00:00 /usr/libexec/gvfsd-service
root          4879    3052  0 09:50 ?        00:00:00 /usr/libexec/gvfsd-trash --spawner /1:21 /org/gtk/gvfs/nautilus-spaw/0
root          4901    3489  0 09:50 ?        00:00:00 /usr/libexec/gsd-session
root          4974        2  0 09:50 ?        00:00:00 [kworker/0:0ext-rx-con]
root          4977        2  0 09:50 ?        00:00:00 [kworker/0:0ext-rx-con]
root          4990    3927  0 09:50 ?        00:00:00 /usr/bin/gnome-shell-x11-frames
root          5004    3489  0 09:50 ?        00:00:00 /usr/bin/ibus-client --clipboard
root          5005    3489  0 09:50 ?        00:00:00 /usr/bin/ibus-client --clipboard
root          5022    3489  0 09:50 ?        00:00:00 /usr/libexec/ibus-x11
root          5043    3489  0 09:50 ?        00:00:00 /usr/bin/ibus-client --vmpa-session
root          5045    3489  0 09:50 ?        00:00:00 /usr/bin/ibus-client --vmpa-session
root          5072    3927  0 09:50 ?        00:00:00 /usr/bin/gnome-shell/extensions/dingbastersoft.com/app/ding.js -E P /usr/share/gnome-shell/extensions/dingbastersoft.com/app
root          5117    3489  0 09:50 ?        00:00:00 /usr/libexec/nautilus-desktop-gtk
root          5317    3489  0 09:50 ?        00:00:00 /usr/bin/physics -application-service
root          5324    5317  0 09:50 pts/0 00:00:00 /usr/bin/bash
root          5420        2  0 10:05 ?        00:00:00 [kworker/0:0rcu_gp_w]
root          5455        2  0 10:00 ?        00:00:00 [kworker/0:0rcu_perq_w]
root          5470        2  0 10:00 ?        00:00:00 [kworker/0:0events]
root          5477    5377  0 10:04 pts/0 00:00:00 ps -ef

```

16. Kill a running process.

- ➔ To kill any process we use the command **kill -9**
- ➔ it will suddenly and forcefully stops the process which may affect on configuration.
- ➔ **kill -9 PID**
- ➔ We can also use **kill PID**

17. Install a package using the package manager (e.g., apt or yum).

- ➔ Apt is a package manager which we use in ubuntu
- ➔ To install package we use “**apt install packagename**”
- ➔ But we should have extra privileges to perform this command, so we use **sudo**
- ➔ Eg: **sudo apt install nginx**

```
poojith@Linux: ~/Desktop
poojith@Linux:~/Desktop$ apt install nginx
Error: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)
Error: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?
poojith@Linux:~/Desktop$ sudo apt install nginx
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14 linux-tools-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap nginx-doc

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 90
  Download size: 655 kB
  Space needed: 1,860 kB / 22.5 GB available

Continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 nginx-common all 1.28.3-2ubuntu1.8 [37.7 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 nginx amd64 1.28.3-2ubuntu1.8 [617 kB]
Fetched 655 kB in 2s (315 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database... 193722 files and directories currently installed.)
Preparing to unpack ./nginx-common_1.28.3-2ubuntu1.8_all.deb...
Unpacking nginx-common (1.28.3-2ubuntu1.8)...
Selecting previously unselected package nginx.
Preparing to unpack ./nginx_1.28.3-2ubuntu1.8_amd64.deb...
Unpacking nginx (1.28.3-2ubuntu1.8)...
Setting up nginx-common (1.28.3-2ubuntu1.8)...
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.8)...
  * Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for ufw (0.36.2-9build1)...
poojith@Linux:~/Desktop$
```

18. Update the system packages.

- ➔ To upgrade the system packages we have a command called “**apt upgrade**” which will upgrade all the packages available.
- ➔ If permission denies then use “**sudo**” → **sudo apt upgrade**

```
poojith@Linux:~/desktop$ apt upgrade
Error: Could not open lock file /var/lib/dpkg/lock-frontend - open (13: Permission denied)
Error: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontend), are you root?
poojith@Linux:~/desktop$ sudo apt upgrade
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14 linux-tools-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Upgrading:
alsa-ucm-conf cups-filters-core-drivers gnome-control-center-data libadwaita-1-0 libgtk-4-1 libpipewire-4.0-3 linux-firmware-qlogic
apport fwupd gnome-control-center-faces libatk-adaptor libgtk-4-bin linux-firmware-and-graphics linux-firmware-qualcomm-graphics
apport-core-dump-handler gdm3 gnome-initial-setup libatk-bridge2.0-0t64 libgtk-4-common linux-firmware-broadcom linux-firmware-qualcomm-misc
apport-gtk gir1.2-adv-1 gnome-session-bin libatk1.0-0t64 libinput-bin linux-firmware-broadcom-wireless linux-firmware-realtek
at-spi2-common gir1.2-atk-1.0 gnome-session-common libatspi2.0-0t64 libinput10 linux-firmware-intel-graphics localsearch
at-spi2-core gir1.2-atspi-2.0 gnome-shell libbluetooth3 libmutter-0-0 linux-firmware-intel-wireless mutter-common
base-files gir1.2-gdm-1.0 gnome-shell-common libfontconfig1 linux-firmware-marvell-presteria mutter-common-bin
bluez gir1.2-gtk-4.0 gnome-text-editor libfontconfig1 linux-firmware-marvell-wireless orca
bluez-cups gir1.2-mutter-10 gstreamer1.0-packagekit libpackagekit-glib2-10 linux-firmware-mellanox-spectrum packagekit
bluez-obexd gir1.2-packagekitglib-1.0 gtk-update-icon-cache libcrypt20 libpackagekit-glib2-10 linux-firmware-netronome papers
cups-filters gnome-control-center iproute2 libgdm1 libpipewire-4.0-6 linux-firmware-nvidia-graphics papers-common

Installing dependencies:
gst-audio-thumbnailer gst-video-thumbnailer

Not upgrading yet due to phasing:
python3-software-properties software-properties-common

Summary:
Upgrading: 88, Installing: 2, Removing: 0, Not Upgrading: 2
Download size: 617 MB
Space needed: 1,961 kB / 22.5 GB available

Continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 base-files amd64 14ubuntu1 [75.0 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcrypt20 amd64 1.12.0-2ubuntu1 [671 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 iproute2 amd64 6.19.0-1ubuntu1 [1,198 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-ptinial amd64 1.576.2 [8,594 B]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-release-upgrader-gtk all 1:26.04.22 [9,854 B]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-release-upgrader-core all 1:26.04.22 [24.9 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-distupgrader all 1:26.04.22 [162 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-standard amd64 1.576.2 [8,582 B]
Err:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 alsa-ucm-conf all 1.2.15.3-1ubuntu1.4
404 Not Found [IP: 91.189.91.81 80]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-problem-report all 2.34.1-0ubuntu0.1 [17.4 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-apport all 2.34.1-0ubuntu0.1 [96.4 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport-core-dump-handler all 2.34.1-0ubuntu0.1 [9,674 B]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport all 2.34.1-0ubuntu0.1 [83.9 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport-gtk all 2.34.1-0ubuntu0.1 [9,852 B]
Get:15 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 at-spi2-common all 2.60.4-0ubuntu0.1 [6,604 B]
Get:16 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libatk-adaptor amd64 2.60.4-0ubuntu0.1 [6,256 B]
Get:17 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-atk-1.0 amd64 2.60.4-0ubuntu0.1 [23.2 kB]
Get:18 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libatk-bridge2.0-0t64 amd64 2.60.4-0ubuntu0.1 [67.2 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libatk1.0-0t64 amd64 2.60.4-0ubuntu0.1 [53.5 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-atspi-2.0 amd64 2.60.4-0ubuntu0.1 [22.6 kB]
Get:21 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 at-spi2-core amd64 2.60.4-0ubuntu0.1 [59.6 kB]
Get:22 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libatspi2.0-0t64 amd64 2.60.4-0ubuntu0.1 [85.5 kB]
Get:23 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bluez amd64 5.85-4ubuntu0.1 [1,550 kB]
Get:24 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bluez-cups amd64 5.85-4ubuntu0.1 [27.6 kB]
Get:25 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bluez-obexd amd64 5.85-4ubuntu0.1 [244 kB]
Get:26 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cups-filters-core-drivers amd64 2.0.1-0ubuntu4.1 [51.2 kB]
Get:27 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cups-filters amd64 2.0.1-0ubuntu4.1 [125 kB]
Get:28 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 fwupd amd64 2.1.1-1ubuntu3.1 [12.7 MB]
Get:29 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libfwupd3 amd64 2.1.1-1ubuntu3.1 [145 kB]
Err:30 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-control-center-data all 1:50.0-0ubuntu6.1
404 Not Found [IP: 91.189.91.81 80]
Get:31 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libgtk-4-common all 4.22.4+ds-0ubuntu0.1 [1,546 kB]
Get:32 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libgtk-4-1 amd64 4.22.4+ds-0ubuntu0.1 [3,552 kB]
Get:33 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libadwaita-1-0 amd64 1.9.1-0ubuntu0.1 [530 kB]
Get:34 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-initial-setup amd64 50.0-0ubuntu7.1 [898 kB]
Get:35 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-session-bin amd64 50.1-0ubuntu0.1 [85.1 kB]
Get:36 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gdm3 amd64 50.1-0ubuntu0.1 [246 kB]
Get:37 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libgdm1 amd64 50.1-0ubuntu0.1 [60.5 kB]
Get:38 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-gdm-1.0 amd64 50.1-0ubuntu0.1 [10.0 kB]
Get:39 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-mutter-18 amd64 50.1-0ubuntu2.2 [134 kB]
Get:40 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 mutter-common all 50.1-0ubuntu2.2 [38.7 kB]
Get:41 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 mutter-common-bin amd64 50.1-0ubuntu2.2 [54.5 kB]
Err:42 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libinput-bin amd64 1.31.1-1ubuntu1
404 Not Found [IP: 91.189.91.81 80]
Err:43 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libinput10 amd64 1.31.1-1ubuntu1
404 Not Found [IP: 91.189.91.81 80]
Get:44 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libmutter-18-0 amd64 50.1-0ubuntu2.2 [1,498 kB]
Get:45 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libnalccontent-common all 0.14.0-0ubuntu1.1 [11.1 kB]
Get:46 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libnalccontent-0-0 amd64 0.14.0-0ubuntu1.1 [40.1 kB]
Err:47 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-control-center amd64 1:50.0-0ubuntu6.1
404 Not Found [IP: 91.189.91.81 80]
Get:48 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-gtk-4.0 amd64 4.22.4+ds-0ubuntu0.1 [224 kB]
```



```

Get:49 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-adv-1 amd64 1.9.1-0ubuntu1.1 [46.7 kB]
Err:50 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-shell amd64 50.1-0ubuntu1.1
404 Not Found [IP: 91.189.91.81 80]
Err:51 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-shell-common all 50.1-0ubuntu1.1
404 Not Found [IP: 91.189.91.81 80]
Get:52 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-session all 50.1-0ubuntu0.1 [3,750 B]
Get:53 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-session-common all 50.1-0ubuntu1.1 [18.9 kB]
Get:54 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpackagekit-glib2-18 amd64 1.3.4-3ubuntu1.1 [121 kB]
Get:55 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-packagekitglib-1.0 amd64 1.3.4-3ubuntu1.1 [25.5 kB]
Err:56 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-control-center-faces all 1.50.0-0ubuntu1.1
404 Not Found [IP: 91.189.91.81 80]
Get:57 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gnome-text-editor amd64 50.1-0ubuntu0.1 [199 kB]
Get:58 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 gst-audio-thumbnailer amd64 1.0.0-0ubuntu1 [315 kB]
Get:59 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 gst-video-thumbnailer amd64 1.0.0-0ubuntu1 [332 kB]
Get:60 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gstreamer1.0-packagekit amd64 1.3.4-3ubuntu1.1 [9,352 B]
Get:61 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 packagekit amd64 1.3.4-3ubuntu1.1 [505 kB]
Get:62 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gtk-update-icon-cache amd64 4.22.4+ds-0ubuntu0.1 [42.7 kB]
Get:63 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libbluetooth3 amd64 5.85-4ubuntu0.1 [78.0 kB]
Get:64 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libfpprint-2.td1 amd64 1:1.95.1+td1-0ubuntu2 [93.3 kB]
Get:65 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libfpprint-2-2 amd64 1:1.95.1+td1-0ubuntu2 [305 kB]
Get:66 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libgtk-4-bin amd64 4.22.4+ds-0ubuntu1.1 [3,675 kB]
Get:67 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libtcpacess amd64 6.10.1-1ubuntu0.1 [19.0 kB]
Get:68 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 papers-common amd64 50.2-0ubuntu1.1 [2,857 kB]
Get:69 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libppsview-4.0-5 amd64 50.2-0ubuntu1.1 [179 kB]
Get:70 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 papers-common all 50.2-0ubuntu1.1 [115 kB]
Get:71 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libppsdokument-4.0-6 amd64 50.2-0ubuntu1.1 [161 kB]
Get:72 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-and-misc all 20260319.git217cae64-0ubuntu1.1 [584 kB]
Get:73 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-and-graphics all 20260319.git217cae64-0ubuntu1.1 [29.6 MB]
Get:74 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-broadcom-wireless all 20260319.git217cae64-0ubuntu1.1 [11.2 MB]
Get:75 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-intel-graphics all 20260319.git217cae64-0ubuntu1.1 [22.0 MB]
Get:76 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-intel-wireless all 20260319.git217cae64-0ubuntu1.1 [102 MB]
Get:77 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-nvidia-presteria all 20260319.git217cae64-0ubuntu1.2 [32.9 MB]
Get:78 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-nvidia-wireless all 20260319.git217cae64-0ubuntu1.1 [7.434 kB]
Get:79 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-nellnox-spectrum all 20260319.git217cae64-0ubuntu1.1 [61.9 MB]
Get:80 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-netronome all 20260319.git217cae64-0ubuntu1.1 [5,765 kB]
Get:81 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-nvidia-graphics all 20260319.git217cae64-0ubuntu1.1 [109 MB]
Get:82 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-logic all 20260319.git217cae64-0ubuntu1.1 [7,178 kB]
Get:83 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-qualcomm-graphics all 20260319.git217cae64-0ubuntu1.1 [11.0 MB]
Get:84 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-qualcomm-misc all 20260319.git217cae64-0ubuntu1.1 [167 MB]
Get:85 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-realtek all 20260319.git217cae64-0ubuntu1.2 [7,301 kB]
Get:86 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 localsearch amd64 3.11.0-1ubuntu1 [416 kB]
Get:87 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 orca all 50.2-0ubuntu1.1 [612 kB]
Get:88 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 resources amd64 1.0-2-0ubuntu1.1 [1,930 kB]
Get:89 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-desktop-minimal amd64 1.570.2 [9,300 B]
Get:90 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-kernel-accessories amd64 1.570.2 [8,330 B]
Fetched 613 MB in 1min 25s (7,216 kB/s)
E: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/g/alsa-ucm-conf/alsa-ucm-conf-1.2.15-3-1ubuntu1.4_all.deb 404 Not Found [IP: 91.189.91.81 80]
Get:88 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 resources amd64 1.0-2-0ubuntu1.1 [1,930 kB]
Get:89 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-desktop-minimal amd64 1.570.2 [9,300 B]
Get:90 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-kernel-accessories amd64 1.570.2 [8,330 B]
Fetched 613 MB in 1min 25s (7,216 kB/s)
Error: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/g/alsa-ucm-conf/alsa-ucm-conf-1.2.15-3-1ubuntu1.4_all.deb 404 Not Found [IP: 91.189.91.81 80]
Error: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/g/gnome-control-center/gnome-control-center-data-50.0-0ubuntu1.1_all.deb 404 Not Found [IP: 91.189.91.81 80]
Error: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/libl/libinput/libinput-bin-1.91.1-1ubuntu1 amd64.deb 404 Not Found [IP: 91.189.91.81 80]
Error: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/libl/libinput/libinput-bin-1.91.1-1ubuntu1 amd64.deb 404 Not Found [IP: 91.189.91.81 80]
Error: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/g/gnome-control-center/gnome-control-center-50.0-0ubuntu1.1 amd64.deb 404 Not Found [IP: 91.189.91.81 80]
Error: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/g/gnome-shell/gnome-shell-50.1-0ubuntu1.1 amd64.deb 404 Not Found [IP: 91.189.91.81 80]
Error: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/g/gnome-shell/gnome-shell-common-50.1-0ubuntu1.1 all.deb 404 Not Found [IP: 91.189.91.81 80]
Error: Failed to fetch https://in.archive.ubuntu.com/ubuntu/pool/main/g/gnome-control-center/gnome-control-center-faces-50.0-0ubuntu1.1_all.deb 404 Not Found [IP: 91.189.91.81 80]
Error: Unable to fetch some archives, maybe run apt update or try with --fix-missing?
pool@thglux:~/Desktop$

```

19. Create a symbolic link.

- ➔ A **symbolic link (symlink)** is a special type of file that acts like a **shortcut** to another file or directory.

➔ We use **ln -s /original file path /path** to create link

```
poojith@Linux:~/Desktop$ ls
Techie
poojith@Linux:~/Desktop$ cd Techie/
poojith@Linux:~/Desktop/Techie$ ls
poojith@Linux:~/Desktop/Techie$ touch Hi
poojith@Linux:~/Desktop/Techie$ ls
Hi
poojith@Linux:~/Desktop/Techie$ vi Hi
poojith@Linux:~/Desktop/Techie$ pwd
/home/poojith/Desktop/Techie
poojith@Linux:~/Desktop/Techie$ ln -s ^C
poojith@Linux:~/Desktop/Techie$ ln -s /home/poojith/Desktop/Techie/Hi /home/poojith/Desktop/

poojith@Linux:~/Desktop/Techie$ ls
Hi
poojith@Linux:~/Desktop/Techie$ cd ..
poojith@Linux:~/Desktop$ ls
Hi Techie
poojith@Linux:~/Desktop$ vi Hi
poojith@Linux:~/Desktop$
```


20. Search for files using the find command.

- ➔ We use **find** command to search for any file in the system.
- ➔ We have the command as – “ **find / -name file_name**”
- ➔ If we don't know the name of the file but we know the extension of the file which we want to search for then we can use “ **find / -name *.txt** ”
- it will give the list of files having extension “.txt”

```
poojith@Linux:~/Desktop$ find / -name Hi
/home/poojith/Desktop/Techie/Hi
/home/poojith/Desktop/Hi
find: '/lost+found': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-fprintd.service-MYUehy': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-fwupd.service-22tXq7': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-upower.service-IdLgpo': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-colord.service-KPisno': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-power-profiles-daemon.service-Y16wzu': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-ModemManager.service-Whh2PW': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-systemd-logind.service-PzV62D': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-switcheroo-control.service-Xadff3': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-polkit.service-CKrNdy': Permission denied
find: '/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-chrony.service-obT360': Permission denied
find: '/tmp/snap-private-tmp': Permission denied
find: '/var/cache/apparmor/ae1a4050.0': Permission denied
find: '/var/cache/apparmor/147ee081.0': Permission denied
find: '/var/cache/apt/archives/partial': Permission denied
find: '/var/cache/private': Permission denied
find: '/var/cache/ldconfig': Permission denied
find: '/var/cache/cups': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-fwupd.service-DopSZf': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-chrony.service-00DB6m': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-polkit.service-8muuHF': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-switcheroo-control.service-45CaHk': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-fprintd.service-RgjLEg': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-power-profiles-daemon.service-KhbIjq': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-ModemManager.service-3gCRRu': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-systemd-logind.service-l3B5Es': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-colord.service-erxwfj': Permission denied
find: '/var/tmp/systemd-private-553ff09dcc264bbb9ff56c294c1a4a08-upower.service-Xk2v0Y': Permission denied
```

```
find: '/usr/src/obj': Permission denied
poojith@Linux:~/Desktop$ find / -name *.txt
/usr/src/linux-headers-7.0.0-28-generic/scripts/head-object-list.txt
/usr/src/linux-headers-7.0.0-28-generic/scripts/context-analysis-suppression.txt
/usr/src/linux-headers-7.0.0-28-generic/scripts/spelling.txt
/usr/src/linux-headers-7.0.0-14-generic/scripts/head-object-list.txt
/usr/src/linux-headers-7.0.0-14-generic/scripts/context-analysis-suppression.txt
/usr/src/linux-headers-7.0.0-14-generic/scripts/spelling.txt
/usr/src/linux-headers-7.0.0-28/scripts/head-object-list.txt
/usr/src/linux-headers-7.0.0-28/scripts/context-analysis-suppression.txt
/usr/src/linux-headers-7.0.0-28/scripts/spelling.txt
/usr/src/linux-headers-7.0.0-28/arch/sh/include/mach-kfr2r09/mach/partner-jet-setup.txt
/usr/src/linux-headers-7.0.0-28/arch/sh/include/mach-ecovec24/mach/partner-jet-setup.txt
/usr/src/linux-headers-7.0.0-14/scripts/head-object-list.txt
/usr/src/linux-headers-7.0.0-14/scripts/context-analysis-suppression.txt
/usr/src/linux-headers-7.0.0-14/scripts/spelling.txt
/usr/src/linux-headers-7.0.0-14/arch/sh/include/mach-kfr2r09/mach/partner-jet-setup.txt
/usr/src/linux-headers-7.0.0-14/arch/sh/include/mach-ecovec24/mach/partner-jet-setup.txt
/usr/lib/xorg/protocol.txt
/usr/lib/python3/dist-packages/dbus_python-1.4.0.egg-info/dependency_links.txt
/usr/lib/python3/dist-packages/dbus_python-1.4.0.egg-info/top_level.txt
/usr/lib/python3/dist-packages/jaraco_context-6.0.1.dist-info/top_level.txt
/usr/lib/python3/dist-packages/netaddr-1.3.0.dist-info/entry_points.txt
/usr/lib/python3/dist-packages/netaddr-1.3.0.dist-info/top_level.txt
/usr/lib/python3/dist-packages/netaddr/eui/oui.txt
/usr/lib/python3/dist-packages/netaddr/eui/iab.txt
/usr/lib/python3/dist-packages/netaddr/tests/eui/sample_iab.txt
/usr/lib/python3/dist-packages/netaddr/tests/eui/sample_oui.txt
/usr/lib/python3/dist-packages/zip-3.23.0.dist-info/top_level.txt
/usr/lib/python3/dist-packages/lazr_restfulclient-0.14.6.dist-info/namespace_packages.txt
/usr/lib/python3/dist-packages/lazr_restfulclient-0.14.6.dist-info/top_level.txt
/usr/lib/python3/dist-packages/oauthlib-3.3.1.dist-info/top_level.txt
/usr/lib/python3/dist-packages/markdown_it_py-3.0.0.dist-info/entry_points.txt
/usr/lib/python3/dist-packages/distro_info-1.15.egg-info/dependency_links.txt
/usr/lib/python3/dist-packages/distro_info-1.15.egg-info/top_level.txt
/usr/lib/python3/dist-packages/pkg_resources/api_tests.txt
/usr/lib/python3/dist-packages/pkg_resources/tests/data/my-test-package_unpacked-egg/my_test_package-1.0-py3.7.egg/EGG-INFO/SOURCES.txt
/usr/lib/python3/dist-packages/pkg_resources/tests/data/my-test-package_unpacked-egg/my_test_package-1.0-py3.7.egg/EGG-INFO/dependency_links.txt
/usr/lib/python3/dist-packages/pkg_resources/tests/data/my-test-package_unpacked-egg/my_test_package-1.0-py3.7.egg/EGG-INFO/top_level.txt
/usr/lib/python3/dist-packages/ubuntu_pro_client-8001.egg-info/entry_points.txt
/usr/lib/python3/dist-packages/ubuntu_pro_client-8001.egg-info/dependency_links.txt
/usr/lib/python3/dist-packages/ubuntu_pro_client-8001.egg-info/requires.txt
/usr/lib/python3/dist-packages/ubuntu_pro_client-8001.egg-info/top_level.txt
/usr/lib/python3/dist-packages/ufw-0.36.2.egg-info/dependency_links.txt
/usr/lib/python3/dist-packages/ufw-0.36.2.egg-info/top_level.txt
/usr/lib/python3/dist-packages/pillow-12.1.1.egg-info/dependency_links.txt
/usr/lib/python3/dist-packages/pillow-12.1.1.egg-info/requires.txt
```

21. Compress and decompress files using tar.

➔ **Compress File.**

➔ To compress any file we have a command called “**tar**”

➔ “**tar cvf new_filename.tar old_filename**” - will compress the give file.

➔ **Decompress**

➔ To decompress we use command “**tar xvf filename.tar**”

```
poojith@Linux:~/Desktop/Techie$ cd compressfile/
poojith@Linux:~/Desktop/Techie/compressfile$ touch a b c d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$ ls
a b c d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$ cd ..
poojith@Linux:~/Desktop/Techie$ ls
Hi compressfile
poojith@Linux:~/Desktop/Techie$ cd compressfile/
poojith@Linux:~/Desktop/Techie/compressfile$ vi a
poojith@Linux:~/Desktop/Techie/compressfile$ vi c
poojith@Linux:~/Desktop/Techie/compressfile$ ls
a b c d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$ tar cvf compressfile
tar: Cowardly refusing to create an empty archive
Try 'tar --help' or 'tar --usage' for more information.
poojith@Linux:~/Desktop/Techie/compressfile$ cd ..
poojith@Linux:~/Desktop/Techie$ s
s: command not found
poojith@Linux:~/Desktop/Techie$ ls
Hi compressfile
poojith@Linux:~/Desktop/Techie$ tar cvf compressfile/
tar: Cowardly refusing to create an empty archive
Try 'tar --help' or 'tar --usage' for more information.
poojith@Linux:~/Desktop/Techie$ cd compressfile/
poojith@Linux:~/Desktop/Techie/compressfile$ tar cvf a compressfile
tar: compressfile: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
poojith@Linux:~/Desktop/Techie/compressfile$ sudo tar cvf a compressfile
[sudo: authenticate] Password:
tar: compressfile: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
poojith@Linux:~/Desktop/Techie/compressfile$ tar cvf compresseda.tar a
a
poojith@Linux:~/Desktop/Techie/compressfile$ ls
a b c compresseda.tar d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$ rm a
poojith@Linux:~/Desktop/Techie/compressfile$ ls
b c compresseda.tar d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$ tar xvf compresseda.tar
a
poojith@Linux:~/Desktop/Techie/compressfile$ ls
a b c compresseda.tar d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$ rm compresseda.tar
poojith@Linux:~/Desktop/Techie/compressfile$ ls
a b c d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$
```

22. Monitor system resources with top or htop.

➔ We use Top command to get complete details of the cpu utilization.

```
top - 11:29:55 up 1:50, 1 user, load average: 0.19, 0.09, 0.09
Tasks: 293 total, 2 running, 291 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.2 sy, 0.0 ni, 99.5 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 7422.7 total, 3745.3 free, 1544.2 used, 2481.4 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 5878.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3927	poojith	20	0	7312520	448560	188192	S	3.3	5.9	4:08.76	gnome-shell
5317	poojith	20	0	4280228	384556	163136	S	2.6	5.1	1:54.33	ptxis
880	root	20	0	0	0	0	I	0.3	0.0	0:00.97	kworker/u36:5-events_unbound
8267	poojith	20	0	13428	6316	4148	R	0.3	0.1	0:00.03	top
1	root	20	0	27008	17616	11724	S	0.0	0.2	0:13.45	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.34	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slab_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u32:0-qrtz_ns_handler
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.21	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:10.32	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:03.70	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.03	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kprobe-optimizer
20	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
23	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
24	root	rt	0	0	0	0	S	0.0	0.0	0:01.28	migration/1
25	root	20	0	0	0	0	S	0.0	0.0	0:02.32	ksoftirqd/1
28	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/2
29	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/2
30	root	rt	0	0	0	0	S	0.0	0.0	0:00.75	migration/2
31	root	20	0	0	0	0	S	0.0	0.0	0:00.36	ksoftirqd/2
34	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/3
35	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/3
36	root	rt	0	0	0	0	S	0.0	0.0	0:00.67	migration/3
37	root	20	0	0	0	0	S	0.0	0.0	0:00.20	ksoftirqd/3
39	root	0	-20	0	0	0	I	0.0	0.0	0:00.50	kworker/3:0H-kblockd
40	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/4
41	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/4
42	root	rt	0	0	0	0	S	0.0	0.0	0:00.67	migration/4
43	root	20	0	0	0	0	S	0.0	0.0	0:00.18	ksoftirqd/4

➔ htop will perform the same action as **top** but it will provide the output in an interactive way

```
0% 0.0% 2.1% 0.7% 1.22G/7.25G Tasks: 128, 464 thr, 166 kthr: 1 running
0K/0K Load average: 0.31 0.14 0.11
Uptime: 01:52:20
```

PID	USER	PRI	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3927	poojith	20	0	7141M	437M	103M	S	3.4	5.9	1:13:15	/usr/bin/gnome-shell --node-ubuntu
5317	poojith	20	0	4179M	375M	159M	S	2.7	5.1	0:25:02	/usr/bin/ptxis --gapplication-service
4003	poojith	20	0	7140M	437M	0	S	2.0	5.9	0:05:05	/usr/bin/gnome-shell --node-ubuntu
8266	poojith	20	0	1240M	615M	1904	R	2.0	0.1	0:00:17	htop
3884	poojith	20	0	7140M	437M	0	S	1.3	5.9	0:17:30	/usr/bin/gnome-shell --node-ubuntu
3981	poojith	20	0	7140M	437M	0	S	0.7	5.9	0:15:54	/usr/bin/gnome-shell --node-ubuntu
3983	poojith	20	0	7140M	437M	0	S	0.7	5.9	0:16:08	/usr/bin/gnome-shell --node-ubuntu
5340	poojith	20	0	4179M	375M	0	S	0.7	5.1	0:18:01	/usr/bin/ptxis --gapplication-service
5343	poojith	20	0	4179M	375M	0	S	0.7	5.1	0:08:08	/usr/bin/ptxis --gapplication-service
5345	poojith	20	0	4179M	375M	0	S	0.7	5.1	0:09:52	/usr/bin/ptxis --gapplication-service
5346	poojith	20	0	4179M	375M	0	S	0.7	5.1	0:09:29	/usr/bin/ptxis --gapplication-service
5347	poojith	20	0	4179M	375M	0	S	0.7	5.1	0:09:05	/usr/bin/ptxis --gapplication-service
1	root	20	0	27008	17616	11724	S	0.0	0.2	0:13:56	/usr/lib/systemd/systemd --switched-root --system --deserialize=51
602	root	19	-1	50932	10832	10976	S	0.0	0.2	0:02:01	/usr/lib/systemd/systemd-journald
735	systemd-oo	20	0	16806	7780	6732	S	0.0	0.1	0:04:01	/usr/lib/systemd/systemd-oomd
738	systemd-re	20	0	22536	14996	12400	S	0.0	0.2	0:00:79	/usr/lib/systemd/systemd-resolved
747	root	20	0	41448	15172	9280	S	0.0	0.2	0:03:41	/usr/lib/systemd/systemd-udev
1379	avahi	20	0	6740	4790	4416	S	0.0	0.1	0:01:07	avahi-daemon: running [Linux.local]
1383	root	20	0	2080	940	1020	S	0.0	0.0	0:00:19	/bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
1385	messagebus	20	0	11872	8740	4708	S	0.0	0.1	0:12:06	dbus-daemon --system --address=systemd: --nofork --nopidfile --system-activation --syslog-only
1413	root	20	0	49428	38172	14900	S	0.0	0.4	0:01:08	/usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
1415	polkitd	20	0	3998	19136	1792	S	0.0	0.3	0:03:70	/usr/lib/polkit-1/polkitd --no-debug --log-level=notice
1434	root	20	0	2131M	4480	2050	S	0.0	0.6	0:00:04	/usr/lib/napd/napd
1438	root	20	0	324M	11050	12900	S	0.0	0.2	0:01:64	/usr/libexec/accounts-daemon
1441	root	20	0	9660	3352	108	S	0.0	0.0	0:00:13	/usr/sbin/cron -F -P
1447	root	20	0	383M	7944	7088	S	0.0	0.1	0:00:40	/usr/libexec/switcheroo-control
1455	root	20	0	11508	9736	1296	S	0.0	0.1	0:03:15	/usr/lib/systemd/systemd-logind
1457	root	20	0	543M	26692	11044	S	0.0	0.3	0:01:00	/usr/libexec/udisks2/udisksd
1478	avahi	20	0	6500	1636	1332	S	0.0	0.0	0:00:00	avahi-daemon: chroot helper
1492	root	20	0	543M	26692	0	S	0.0	0.3	0:00:48	/usr/libexec/udisks2/udisksd
1493	syslog	20	0	215M	6324	6880	S	0.0	0.1	0:00:28	/usr/sbin/rsyslogd -n -iNONE
1495	root	20	0	543M	26692	0	S	0.0	0.3	0:00:01	/usr/libexec/udisks2/udisksd
1505	root	20	0	2131M	44884	0	S	0.0	0.6	0:03:27	/usr/lib/napd/napd

23. Create and manage user groups.

- ➔ to create a group we use command “**groupadd filename**”
- ➔ if you don't have extra privileges use **sudo**
- ➔ then it will be created
- ➔ check with **getent group groupname** to know if it is created

```
poojith@Linux:~/Desktop/Techie/compressfile$ groups
poojith adm cdrom sudo dip plugdev users lpadmin lxd vboxsf
poojith@Linux:~/Desktop/Techie/compressfile$ groupadd Dev_Eng
groupadd: Permission denied.
groupadd: cannot lock /etc/group; try again later.
poojith@Linux:~/Desktop/Techie/compressfile$ sudo groupadd Dev_Eng
poojith@Linux:~/Desktop/Techie/compressfile$ groups
poojith adm cdrom sudo dip plugdev users lpadmin lxd vboxsf
poojith@Linux:~/Desktop/Techie/compressfile$ ls
a b c d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$ groups
poojith adm cdrom sudo dip plugdev users lpadmin lxd vboxsf
poojith@Linux:~/Desktop/Techie/compressfile$ getent group Dev_Eng
Dev_Eng:x:1002:
poojith@Linux:~/Desktop/Techie/compressfile$
```

➔ add users to the group

➔ **sudo usermod -aG Dev_Eng Poojith** (adding user)

➔ Changed the user/owner of a file named ‘a’

```
poojith@Linux:~/Desktop/Techie/compressfile$ sudo usermod -aG Dev_Eng Ravutla
poojith@Linux:~/Desktop/Techie/compressfile$ groups
poojith adm cdrom sudo dip plugdev users lpadmin lxd vboxsf
poojith@Linux:~/Desktop/Techie/compressfile$ groups poojith
poojith : poojith adm cdrom sudo dip plugdev users lpadmin lxd vboxsf
poojith@Linux:~/Desktop/Techie/compressfile$ groups Ravutla
Ravutla : Ravutla Dev_Eng
poojith@Linux:~/Desktop/Techie/compressfile$ chown Ravutla:Dev_Eng a
chown: changing ownership of 'a': Operation not permitted (os error 1)
poojith@Linux:~/Desktop/Techie/compressfile$ sudo chown Ravutla:Dev_Eng a
poojith@Linux:~/Desktop/Techie/compressfile$ ls -l a
-rw-rw-r-- 1 Ravutla Dev_Eng 10240 Aug  8 11:11 a
poojith@Linux:~/Desktop/Techie/compressfile$ ls
a b c d e f g h
poojith@Linux:~/Desktop/Techie/compressfile$ ls -l b
-rw-rw-r-- 1 poojith poojith 0 Aug  8 10:58 b
poojith@Linux:~/Desktop/Techie/compressfile$ ls -l
total 16
-rw-rw-r-- 1 Ravutla Dev_Eng 10240 Aug  8 11:11 a
-rw-rw-r-- 1 poojith poojith 0 Aug  8 10:58 b
-rw-rw-r-- 1 poojith poojith 92 Aug  8 11:08 c
-rw-rw-r-- 1 poojith poojith 0 Aug  8 10:58 d
-rw-rw-r-- 1 poojith poojith 0 Aug  8 10:58 e
-rw-rw-r-- 1 poojith poojith 0 Aug  8 10:58 f
-rw-rw-r-- 1 poojith poojith 0 Aug  8 10:58 g
-rw-rw-r-- 1 poojith poojith 0 Aug  8 10:58 h
poojith@Linux:~/Desktop/Techie/compressfile$
```

24. Set up SSH password less authentication.

Objective

To configure SSH key-based authentication without entering a password each time.

Purpose

SSH keys provide convenient and generally stronger authentication for server administration and automation.

Implementation

On the client machine:

```
ssh-keygen -t ed25519
```

Copy the public key to the server:

```
ssh-copy-id user@SERVER_IP
```

Connect:

```
ssh user@SERVER_IP
```

Protect the private key:

```
chmod 700 ~/.ssh/id_ed25519
```

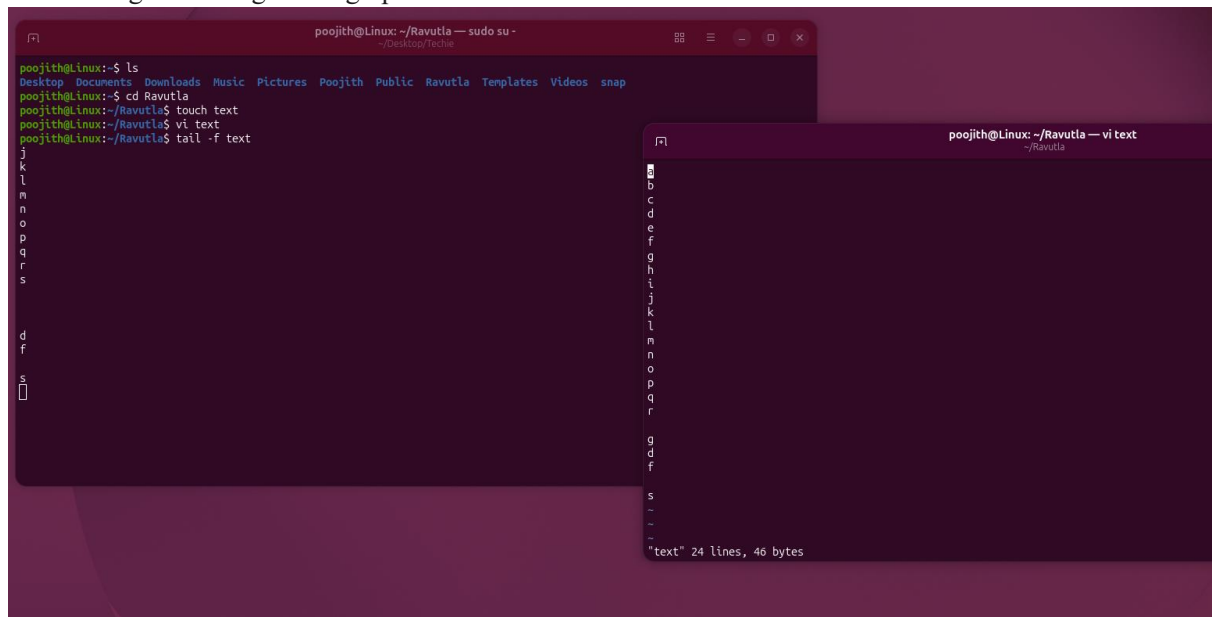
SSH key-based passwordless authentication was successfully configured

```
pooji@Poojith MINGW64 ~/Downloads
$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/pooji/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/pooji/.ssh/id_ed25519
Your public key has been saved in /c/Users/pooji/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:USwJIMJVP29Chj7thQVopiMlIIAGWovFcYicJSaaU pooji@Poojith
The key's randomart image is:
+--[ED25519 256]--+
|O*@=*=.o o.|
|BBB*+ . +..|
|**Eoo. ...|
|.. . .o ..|
|.... .S.|
|. .+ . .|
|.o o .|
| o.. .|
|.o..|
+-----[SHA256]-----+

pooji@Poojith MINGW64 ~/Downloads
$ ls -a ~/.ssh
./ ../ id_ed25519 id_ed25519.pub id_rsa id_rsa.pub known_hosts known_hosts.old

pooji@Poojith MINGW64 ~/Downloads
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIA07gSg9YjUyLBbBseXXDTxc8l6RzciLWqvEv9zc2LuT pooji@Poojith
```


25. Monitor log files using tail or grep.

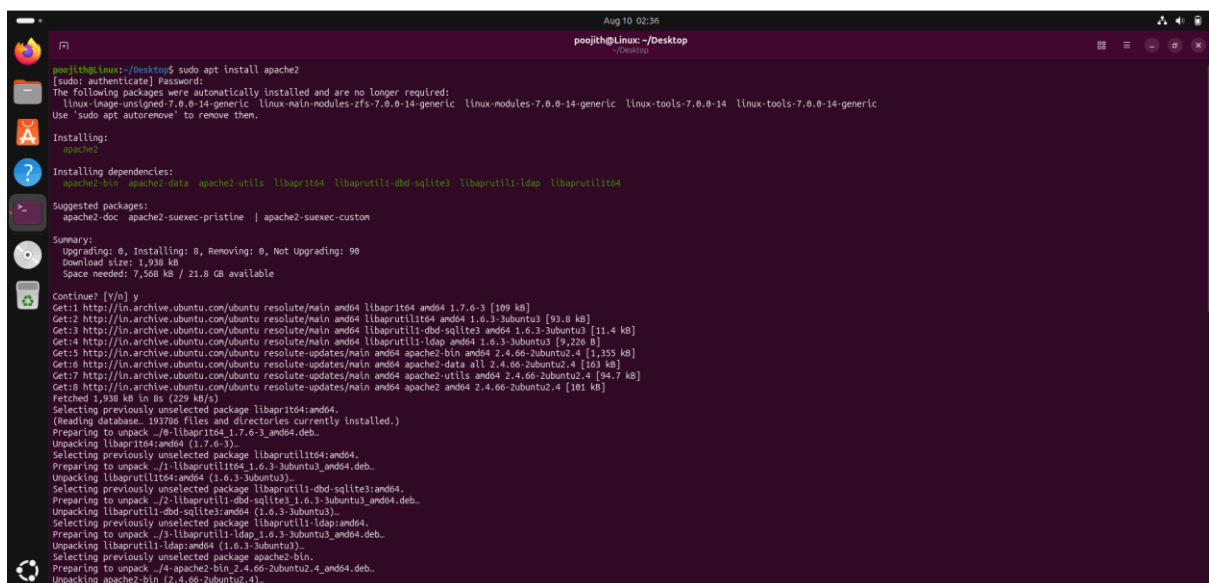


26. Set up a web server (e.g., Apache or Nginx).

Task	Command
Install Apache	<code>sudo apt install apache2 -y</code>
Start	<code>sudo systemctl start apache2</code>
Stop	<code>sudo systemctl stop apache2</code>

→ as previously nginx is already running and apache2 and nginx will use the same port number:

80. I have stopped nginx and then started apache2




```

poojith@Linux:~/Desktop$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Mon 2020-08-10 02:36:56 UTC; 57s ago
  Invocation: 22a1266dcd34ba4b8a892baa3dbb882
     Docs: https://httpd.apache.org/docs/2.4/
    Process: 7250 ExecStart=/usr/sbin/apachectl start (code=exited, status=1/FAILURE)
 Main PID: 7250 (code=exited, status=1/FAILURE)
   Status: "Heading configuration..."
  Men Peak: 2.3M
    CPU: 52ms

Aug 10 02:36:56 Linux systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 10 02:36:56 Linux apachectl[7250]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Aug 10 02:36:56 Linux apachectl[7250]: (98)Address already in use: AH00072: make_sock: could not bind to address [::]:80
Aug 10 02:36:56 Linux apachectl[7250]: (98)Address already in use: AH00072: make_sock: could not bind to address 0.0.0.0:80
Aug 10 02:36:56 Linux apachectl[7250]: no listening sockets available, shutting down
Aug 10 02:36:56 Linux apachectl[7250]: AH00015: Unable to open logs
Aug 10 02:36:56 Linux systemd[1]: apache2.service: Main process exited, code=exited, status=1/FAILURE
Aug 10 02:36:56 Linux systemd[1]: apache2.service: Failed with result 'exit-code'.
Aug 10 02:36:56 Linux systemd[1]: Failed to start apache2.service - The Apache HTTP Server.
poojith@Linux:~/Desktop$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2020-08-10 02:29:52 UTC; 6min ago
  Invocation: 24e846ef28446b562840247d7de25
     Docs: man:nginx(8)
 Main PID: 2277 (nginx)
    Tasks: 9 (limit: 6354)
  Memory: 9M (peak: 10.1M)
    CPU: 26ms
  CGroup: /system.slice/nginx.service
          └─2277 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
             └─2278 "nginx: worker process"
             └─2279 "nginx: worker process"
             └─2280 "nginx: worker process"
             └─2281 "nginx: worker process"
             └─2282 "nginx: worker process"
             └─2283 "nginx: worker process"
             └─2284 "nginx: worker process"
             └─2285 "nginx: worker process"

Aug 10 02:29:51 Linux systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Aug 10 02:29:52 Linux systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
poojith@Linux:~/Desktop$ sudo systemctl stop nginx
poojith@Linux:~/Desktop$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2020-08-10 02:38:55 UTC; 4s ago
  Duration: 9min 3.102s
  Invocation: 24e846ef28446b562840247d7de25
     Docs: man:nginx(8)
    Process: 7312 ExecStop=/sbin/start-stop-daemon --quiet --stop --retry QUIT/5 --pidfile /run/nginx.pid (code=exited, status=0/SUCCESS)
 Main PID: 2277 (code=exited, status=0/SUCCESS)
  Men Peak: 12.4M
    CPU: 305ms

Aug 10 02:29:51 Linux systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Aug 10 02:29:52 Linux systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server...
Aug 10 02:38:55 Linux systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
Aug 10 02:38:55 Linux systemd[1]: nginx.service: Deactivated successfully.
Aug 10 02:38:55 Linux systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
poojith@Linux:~/Desktop$ sudo systemctl start apache2
poojith@Linux:~/Desktop$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2020-08-10 02:39:07 UTC; 5s ago
  Invocation: 1762399e1d849f0bab87defb82d59f
     Docs: https://httpd.apache.org/docs/2.4/
 Main PID: 7208 (apache2)
   Status: "Processing requests..."
    Tasks: 55 (limit: 6354)
  Memory: 6.3M (peak: 6.9M)
    CPU: 92ms
  CGroup: /system.slice/apache2.service
          └─7208 /usr/sbin/apache2 -k start -DFOREGROUND
             └─7232 /usr/sbin/apache2 -k start -DFOREGROUND
             └─7233 /usr/sbin/apache2 -k start -DFOREGROUND

Aug 10 02:39:07 Linux systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 10 02:39:07 Linux apachectl[7208]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Aug 10 02:39:07 Linux systemd[1]: Started apache2.service - The Apache HTTP Server.
poojith@Linux:~/Desktop$

```

27. Configure and secure a MySQL Database.

Install	<code>sudo apt install mysql-server -y</code>
Check service	<code>sudo systemctl status mysql</code>
Start	<code>sudo systemctl start mysql</code>
Enable at boot	<code>sudo systemctl enable mysql</code>
Secure MySQL	<code>sudo mysql_secure_installation</code>

```
poojitha@linux:~/Desktop$ sudo apt install mysql-server
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.6.0-14-generic linux-main-modules-zfs-7.6.0-14-generic linux-modules-7.6.0-14-generic linux-tools-7.6.0-14 linux-tools-7.6.0-14-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  mysql-server

Installing dependencies:
  libaisi164 libbgs-pm-perl libbgs-perl libgoogle-perftools4t64 libmecab2 libnetllor-minimal4t64 mysql-client-core mysql-server-core
  libbgs-fast-perl libbgs-bin libbgs4t64 libhtml-template-perl libprotobuf-lite32t64 mysql-client mysql-common

Suggested packages:
  libipc-sharedcache-perl

Recommended packages:
  mecab-ipadic-utf

Summary:
  Upgrading: 0, Installing: 16, Removing: 0, Not Upgrading: 98
  Download size: 24.3 MB
  Space needed: 182 MB / 21.8 GB available

Continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 mysql-common all 5.8+1.1.1ubuntu2 [7,882 B]
Get:2 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libtcmalloc-minimal4t64 amd64 2.18.1-1 [138 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libgoogle-perftools4t64 amd64 2.18.1-1 [199 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 mysql-client-core amd64 8.4.10-0ubuntu26.04.1 [2,288 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 mysql-client amd64 8.4.10-0ubuntu26.04.1 [9,124 B]
Get:6 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libaisi164 amd64 0.3.113-1build1 [7,356 B]
Get:7 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libmecab2 amd64 0.996-15.1build1 [205 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libprotobuf-lite32t64 amd64 3.21.12-15ubuntu1 [251 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 mysql-server-core amd64 8.4.10-0ubuntu26.04.1 [19.4 MB]
Get:10 http://in.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 mysql-server amd64 8.4.10-0ubuntu26.04.1 [1,595 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libbgs-pm-perl all 4.71-1build1 [185 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libbgs4t64 amd64 2.4.5-0.1build1 [27.9 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libbgs-perl amd64 0.0245-1build1 [21.7 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libbgs-fast-perl all 1:2.17-1 [10.3 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libbgs-bin amd64 2.4.5-0.1build1 [12.2 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu/resolute/main amd64 libhtml-template-perl all 2.97-2build1 [58.3 kB]
Fetched 24.3 MB in 11s (2,126 kB/s)
Preconfiguring packages ...
Selecting previously unselected package mysql-common.
(Reading database ... 39494 files and directories currently installed.)
Preparing to unpack ./8-mysql-common-5.8+1.1.1ubuntu2_all.deb.
Unpacking mysql-common (5.8+1.1.1ubuntu2) ...
Processing triggers for libc-bin (2.40-ubuntu2) ...
Processing triggers for libnss-systemd (2.6.18-1ubuntu1) ...
poojitha@linux:~/Desktop$ sudo systemctl start mysql
poojitha@linux:~/Desktop$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-10 02:41:52 UTC; 1min 22s ago
   Invocation: 77d59c4fcaaa402ba75d09e7e0d1d5e9
   Process: 7994 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
   Main PID: 8803 (mysqld)
   Status: "Server is operational"
   Tasks: 34 (limit: 6554)
   Memory: 480.6M (peak: 481.7M)
   CPU: 3.851s
   CGroup: /system.slice/mysql.service
           └─8803 /usr/sbin/mysqld

Aug 10 02:41:50 Linux systemd[1]: Starting mysql.service - MySQL Community Server...
Aug 10 02:41:52 Linux systemd[1]: Started mysql.service - MySQL Community Server.
poojitha@linux:~/Desktop$ sudo systemctl enable mysql
poojitha@linux:~/Desktop$ sudo mysql_secure_installation
mysql Ver 8.4.10-0ubuntu26.04.1 for Linux on x86_64 ((Ubuntu))
Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Usage: mysql [OPTIONS] [database]
  -?, --help            Display this help and exit.
  -I, --help            Synonym for -?.
  --auto-rehash         Enable automatic rehashing. One doesn't need to use
                        'rehash' to get table and field completion, but startup
                        and reconnecting may take a longer time. Disable with
                        --disable-auto-rehash.
                        (Defaults to on; use --skip-auto-rehash to disable.)
  -A, --no-auto-rehash No automatic rehashing. One has to use 'rehash' to get
                        table and field completion. This gives a quicker start of
                        mysql and disables rehashing on reconnect.
  --auto-vertical-output Automatically switch to vertical output mode if the
                        result is wider than the terminal width.
  -B, --batch           Don't use history file. Disable interactive behavior.
                        (Enables --silent.)
  --bind-address=IP     IP address to bind to.
  --binary-as-hex       print binary data as hex. Enabled by default for
                        interactive terminals.
```

28. Set up a Application Server (e.g.,Apache Tomcat)

Purpose

Command

Update package information `sudo apt update`

Install Java (Tomcat requirement) `sudo apt install openjdk-17-jdk -y`

Verify Java installation `java -version`

Download Tomcat `sudo apt install tomcat11`

Check whether Tomcat is running

```
ps -ef | grep -i tomcat
```

```
Check Tomcat port          sudo ss -ltnp | grep 8080
```

Search in browser Localhost:8080

```
pos@l94kmuir:~/kawaii$ sudo apt install openjdk-17-jdk
[sudo: authenticating] Password:
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14 linux-tools-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Installing:
openjdk-17-jdk

Installing dependencies:
ca-certificates-java java-common libatk-wrapper-java-gtk libasn1-dev libbsd-dev libbcb-doc libbcb-dev openjdk-17-jdk-headless openjdk-17-jre-headless x11proto-dev xtrans-dev
fonts-dejavu-extra libatk-wrapper-java libtiff-dev libx11-dev libxcb-doc libxcb-dev openjdk-17-jre uid-dev xorg-sgml-doctools

Suggested packages:
default-jre libice-doc libsm-doc libxft-doc libxkb-dbg libxt-doc openjdk-17-demo openjdk-17-source visusalvm fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhe | fonts-wqy-zenhei fonts-indic

Summary:
Upgrading: 0, Installing: 20, Removing: 0, Not Upgrading: 90
Download size: 127 MB
Space needed: 291 MB / 21.3 GB available

Continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 ca-certificates-java all 20260831 [11.3 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 fonts-dejavu-extra all 2.37-1build1 [1,944 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 java-common all 0.77 [6,682 B]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libatk-wrapper-java-gtk all 0.44.0-1 [55.2 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libatk-wrapper-java-gtk all 0.44.0-1 [48.0 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 xorg-sgml-doctools all 1:1.11-1build1 [10.3 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 x11proto-dev all 2025.1+1 [688 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libice-dev amd64 2:1.1-1build1 [53.5 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 uid-dev amd64 2.41.3-1ubuntu2 [40.5 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libsm-dev amd64 2:1.2.6-1build1 [18.8 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libxau-dev amd64 1:1.0.11-1build2 [9,712 B]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libxkb-dbg amd64 1:1.15-2 [25.7 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 xtrans-dev all 1.0.0-1build1 [66.2 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libxcb-dev amd64 1.17.0-2ubuntu1 [86.3 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libx11-dev amd64 2:1.8.13-1 [739 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libtiff-dev amd64 4:4.5.1-2-1build1 [396 kB]
Get:17 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jre-headless amd64 17.0.19+10-1.26.04.2 [48.1 MB]
Get:18 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jre amd64 17.0.19+10-1.26.04.2 [228 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jdk-headless amd64 17.0.19+10-1.26.04.2 [71.8 MB]
Get:20 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 openjdk-17-jdk amd64 17.0.19+10-1.26.04.2 [2,481 kB]
fetched 127 MB in 55s (5,929 kBs)
Selecting previously unselected package ca-certificates-java.
(Reading database... 194822 files and directories currently installed.)
Preparing to unpack .../ca-certificates-java_20260831_all.deb...
dpkg-query -W -f='${Package} ${Version} ${Architecture}\n'
```

```
Aug 10 07:26
root@kali: ~# cat /etc/apt/sources.list.d/poijithLinux.sources
deb http://ppa.launchpad.net/poijithlinux/+archive/ubuntu/ppa/ubuntu focal main
deb-src http://ppa.launchpad.net/poijithlinux/+archive/ubuntu/ppa/ubuntu focal main
deb http://ppa.launchpad.net/poijithlinux/+archive/ubuntu/ppa/ubuntu focal-updates main
deb-src http://ppa.launchpad.net/poijithlinux/+archive/ubuntu/ppa/ubuntu focal-updates main
deb http://ppa.launchpad.net/poijithlinux/+archive/ubuntu/ppa/ubuntu focal-security main
deb-src http://ppa.launchpad.net/poijithlinux/+archive/ubuntu/ppa/ubuntu focal-security main

Setting up poijithlinux-ppa (17.0.19-18~26.04.2).
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/javac to provide /usr/bin/javac (javac) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jar to provide /usr/bin/jar (jar) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jarsigner to provide /usr/bin/jarsigner (jarsigner) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/javadoc to provide /usr/bin/javadoc (javadoc) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/javap to provide /usr/bin/javap (javap) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jcnd to provide /usr/bin/jcnd (jcnd) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jdb to provide /usr/bin/jdb (jdb) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jdepscan to provide /usr/bin/jdepscan (jdepscan) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jdeps to provide /usr/bin/jdeps (jdeps) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jfr to provide /usr/bin/jfr (jfr) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jinfo to provide /usr/bin/jinfo (jinfo) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jlink to provide /usr/bin/jlink (jlink) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jmap to provide /usr/bin/jmap (jmap) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jmod to provide /usr/bin/jmod (jmod) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jps to provide /usr/bin/jps (jps) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jrunscript to provide /usr/bin/jrunscript (jrunscript) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jschell to provide /usr/bin/jschell (jschell) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstack to provide /usr/bin/jstack (jstack) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstat to provide /usr/bin/jstat (jstat) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jshdb to provide /usr/bin/jshdb (jshdb) in auto mode
Setting up openjdk-17-jre-headless:amd64 (17.0.19+18-1~26.04.2).
poijithLinux: ~#
```



29. create a service file for Apache Tomcat.(Should execute by using systemctl command)

→ Create the service file

sudo nano /etc/systemd/system/tomcat.service

→

Reload systemd sudo systemctl daemon-reload

Start Tomcat sudo systemctl start tomcat

Check status sudo systemctl status tomcat

Enable Tomcat at boot sudo systemctl enable tomcat

Restart Tomcat sudo systemctl restart tomcat

Stop Tomcat sudo systemctl stop tomcat

```
poojith@Linux:~$ cat /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat 11 Custom Service
After=network.target

[Service]
Type=simple
User=tomcat
Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64"
Environment="CATALINA_HOME=/usr/share/tomcat11"
Environment="CATALINA_BASE=/var/lib/tomcat11"
Environment="CATALINA_TMPDIR=/tmp"

ExecStart=/usr/share/tomcat11/bin/catalina.sh run
ExecStop=/usr/share/tomcat11/bin/catalina.sh stop

Restart=on-failure

[Install]
WantedBy=multi-user.target
poojith@Linux:~$
```

```

poojit@kali:~$ sudo systemctl stop tomcat11
poojit@kali:~$ sudo systemctl disable tomcat11
poojit@kali:~$ sudo systemctl is-enabled tomcat11
disabled
poojit@kali:~$ sudo systemctl status tomcat11
○ tomcat11.service - Apache Tomcat 11 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat11.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: https://tomcat.apache.org/tomcat-11.0-doc/index.html

Aug 10 16:01:50 Linux tomcat11[1820]: Starting ProtocolHandler ["http-nio-8080"]
Aug 10 16:01:50 Linux tomcat11[1820]: Server startup in [5520] milliseconds
Aug 10 19:26:35 Linux systemd[1]: Stopping tomcat11.service - Apache Tomcat 11 Web Application Server...
Aug 10 19:26:35 Linux tomcat11[1820]: Pausing ProtocolHandler ["http-nio-8080"]
Aug 10 19:26:35 Linux tomcat11[1820]: Stopping service [Catalina]
Aug 10 19:26:35 Linux tomcat11[1820]: Stopping ProtocolHandler ["http-nio-8080"]
Aug 10 19:26:35 Linux tomcat11[1820]: Destroying ProtocolHandler ["http-nio-8080"]
Aug 10 19:26:35 Linux systemd[1]: tomcat11.service: Deactivated successfully.
Aug 10 19:26:35 Linux systemd[1]: Stopped tomcat11.service - Apache Tomcat 11 Web Application Server.
Aug 10 19:26:35 Linux systemd[1]: tomcat11.service: Consumed 4min 22.079s CPU time over 3h 26min 41.351s wall clock time, 416.5M memory peak.
poojit@kali:~$

```

```

poojit@kali:~$ sudo systemctl daemon-reload
poojit@kali:~$ sudo systemctl start tomcat
poojit@kali:~$ sudo systemctl status tomcat
● tomcat.service - Apache Tomcat 11 Custom Service
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: enabled)
   Active: active (running) since Mon 2026-08-10 20:11:11 UTC; 21s ago
     Invocation: 596fe2b0ad1f4a0b03895a8d2a510f85
    Main PID: 8678 (java)
      Tasks: 42 (limit: 6354)
     Memory: 228.5M (peak: 235.5M)
        CPU: 8.614s
    CGroup: /system.slice/tomcat.service
           └─usr/lib/jvm/java-17-openjdk-and64/bin/java -Djava.util.logging.config.file=/var/lib/tomcat11/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager

Aug 10 20:11:13 Linux catalina.sh[8678]: Deployment of deployment descriptor [/etc/tomcat11/catalina/localhost/host-manager.xml] has finished in [1,046] ms
Aug 10 20:11:13 Linux catalina.sh[8678]: Deploying deployment descriptor [/etc/tomcat11/catalina/localhost/manager.xml]
Aug 10 20:11:13 Linux catalina.sh[8678]: The path attribute with value [/manager] in deployment descriptor [/etc/tomcat11/catalina/localhost/manager.xml] has been ignored
Aug 10 20:11:14 Linux catalina.sh[8678]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JARs that were scanned but no TLDs were
Aug 10 20:11:14 Linux catalina.sh[8678]: Deployment of deployment descriptor [/etc/tomcat11/catalina/localhost/manager.xml] has finished in [507] ms
Aug 10 20:11:14 Linux catalina.sh[8678]: Deploying web application directory [/var/lib/tomcat11/webapps/ROOT]
Aug 10 20:11:14 Linux catalina.sh[8678]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JARs that were scanned but no TLDs were
Aug 10 20:11:14 Linux catalina.sh[8678]: Deployment of web application directory [/var/lib/tomcat11/webapps/ROOT] has finished in [406] ms
Aug 10 20:11:14 Linux catalina.sh[8678]: Starting ProtocolHandler ["http-nio-8080"]
Aug 10 20:11:14 Linux catalina.sh[8678]: Server startup in [2222] milliseconds
poojit@kali:~$ sudo systemctl enable tomcat
Created symlink /etc/systemd/system/multi-user.target.wants/tomcat.service → /etc/systemd/system/tomcat.service.
poojit@kali:~$ sudo systemctl is-enabled tomcat
enabled
poojit@kali:~$

```

```

poojit@kali:~$ sudo systemctl enable tomcat
Created symlink /etc/systemd/system/multi-user.target.wants/tomcat.service → /etc/systemd/system/tomcat.service.
poojit@kali:~$ sudo systemctl is-enabled tomcat
enabled
poojit@kali:~$ sudo systemctl stop tomcat
poojit@kali:~$ sudo systemctl status tomcat
○ tomcat.service - Apache Tomcat 11 Custom Service
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2026-08-10 20:15:22 UTC; 8s ago
     Duration: 4min 9.888s
     Invocation: 596fe2b0ad1f4a0b03895a8d2a510f85
    Process: 8678 ExecStart=/usr/share/tomcat11/bin/catalina.sh run (code=exited, status=143)
    Process: 8932 ExecStop=/usr/share/tomcat11/bin/catalina.sh stop (code=exited, status=0/SUCCESS)
    Main PID: 8678 (code=exited, status=143)
     Mem peak: 253.7M
        CPU: 11.116s

Aug 10 20:15:21 Linux systemd[1]: Stopping tomcat.service - Apache Tomcat 11 Custom Service...
Aug 10 20:15:22 Linux catalina.sh[8937]: Aug 10, 2026 8:15:22 PM org.apache.catalina.startup.Catalina stopServer
Aug 10 20:15:22 Linux catalina.sh[8937]: SEVERE: No shutdown port configured. Shut down server through OS signal. Server not shut down.
Aug 10 20:15:22 Linux catalina.sh[8678]: Pausing ProtocolHandler ["http-nio-8080"]
Aug 10 20:15:22 Linux catalina.sh[8678]: Stopping service [Catalina]
Aug 10 20:15:22 Linux catalina.sh[8678]: Stopping ProtocolHandler ["http-nio-8080"]
Aug 10 20:15:22 Linux catalina.sh[8678]: Destroying ProtocolHandler ["http-nio-8080"]
Aug 10 20:15:22 Linux systemd[1]: tomcat.service: Deactivated successfully.
Aug 10 20:15:22 Linux systemd[1]: Stopped tomcat.service - Apache Tomcat 11 Custom Service.
Aug 10 20:15:22 Linux systemd[1]: tomcat.service: Consumed 11.116s CPU time over 4min 10.347s wall clock time, 253.7M memory peak.

```

```

poojit@kali:~$ sudo systemctl status tomcat
● tomcat.service - Apache Tomcat 11 Custom Service
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-10 20:15:22 UTC; 8s ago
     Duration: 4min 9.888s
     Invocation: 596fe2b0ad1f4a0b03895a8d2a510f85
    Process: 8678 ExecStart=/usr/share/tomcat11/bin/catalina.sh run (code=exited, status=143)
    Process: 8932 ExecStop=/usr/share/tomcat11/bin/catalina.sh stop (code=exited, status=0/SUCCESS)
    Main PID: 8678 (code=exited, status=143)
     Mem peak: 253.7M
        CPU: 11.116s

Aug 10 20:15:21 Linux systemd[1]: Stopping tomcat.service - Apache Tomcat 11 Custom Service...
Aug 10 20:15:22 Linux catalina.sh[8937]: Aug 10, 2026 8:15:22 PM org.apache.catalina.startup.Catalina stopServer
Aug 10 20:15:22 Linux catalina.sh[8937]: SEVERE: No shutdown port configured. Shut down server through OS signal. Server not shut down.
Aug 10 20:15:22 Linux catalina.sh[8678]: Pausing ProtocolHandler ["http-nio-8080"]
Aug 10 20:15:22 Linux catalina.sh[8678]: Stopping service [Catalina]
Aug 10 20:15:22 Linux catalina.sh[8678]: Stopping ProtocolHandler ["http-nio-8080"]
Aug 10 20:15:22 Linux catalina.sh[8678]: Destroying ProtocolHandler ["http-nio-8080"]
Aug 10 20:15:22 Linux systemd[1]: tomcat.service: Deactivated successfully.
Aug 10 20:15:22 Linux systemd[1]: Stopped tomcat.service - Apache Tomcat 11 Custom Service.
Aug 10 20:15:22 Linux systemd[1]: tomcat.service: Consumed 11.116s CPU time over 4min 10.347s wall clock time, 253.7M memory peak.
poojit@kali:~$ sudo systemctl start tomcat
poojit@kali:~$ sudo systemctl status tomcat
● tomcat.service - Apache Tomcat 11 Custom Service
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-10 20:16:26 UTC; 2s ago
     Invocation: f00a5a0b03d34f2aad78f5739a2781be
    Main PID: 8981 (java)
      Tasks: 31 (limit: 6354)
     Memory: 227M (peak: 227M)
        CPU: 8.836s
    CGroup: /system.slice/tomcat.service
           └─9981 /usr/lib/jvm/java-17-openjdk-and64/bin/java -Djava.util.logging.config.file=/var/lib/tomcat11/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager

Aug 10 20:16:27 Linux catalina.sh[8981]: Initializing ProtocolHandler ["http-nio-8080"]
Aug 10 20:16:27 Linux catalina.sh[8981]: Server initialization in [595] milliseconds
Aug 10 20:16:27 Linux catalina.sh[8981]: Starting service [Catalina]
Aug 10 20:16:27 Linux catalina.sh[8981]: Starting Servlet engine: [Apache Tomcat/11.0.22 (Ubuntu)]
Aug 10 20:16:27 Linux catalina.sh[8981]: Deploying deployment descriptor [/etc/tomcat11/catalina/localhost/host-manager.xml]
Aug 10 20:16:27 Linux catalina.sh[8981]: The path attribute with value [/host-manager] in deployment descriptor [/etc/tomcat11/catalina/localhost/host-manager.xml] has been ignored
Aug 10 20:16:28 Linux catalina.sh[8981]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JARs that were scanned but no TLDs were
Aug 10 20:16:29 Linux catalina.sh[8981]: Deployment of deployment descriptor [/etc/tomcat11/catalina/localhost/host-manager.xml] has finished in [1,495] ms
Aug 10 20:16:29 Linux catalina.sh[8981]: Deploying deployment descriptor [/etc/tomcat11/catalina/localhost/manager.xml]
Aug 10 20:16:29 Linux catalina.sh[8981]: The path attribute with value [/manager] in deployment descriptor [/etc/tomcat11/catalina/localhost/manager.xml] has been ignored
[Lines 1-21/21 (END)]

```


30. Print specific columns from a delimited file.

→ Print column 1 `cut -d',' -f1 file`

```
poojith@Linux:~/Desktop/Techie$ ls
Hello Hi Linux New.tar alphabet compressfile delimfile
poojith@Linux:~/Desktop/Techie$ cut -d',' -f2 delimfile
Poojith
Ravutla
Mailaram
Sirikonda
Nizamabad
poojith@Linux:~/Desktop/Techie$ cat delimfile
100,Poojith,ABC
101,Ravutla,DEF
102,Mailaram,GHI
103,Sirikonda,JKL
104,Nizamabad,MNO
poojith@Linux:~/Desktop/Techie$
```

31. Filter and print lines based on a specific pattern or condition.

Find a pattern `grep "pattern" file`

Ignore case `grep -i "pattern" file`

Filter a specific column `awk '$2 == "a"' file`

```
poojith@Linux:~/Desktop$ ls
Hi Techie
poojith@Linux:~/Desktop$ cd Techie/
poojith@Linux:~/Desktop/Techie$ ls
Hi New.tar compressfile
poojith@Linux:~/Desktop/Techie$ vi Hello
poojith@Linux:~/Desktop/Techie$ ls
Hello Hi New.tar compressfile
poojith@Linux:~/Desktop/Techie$ grep a Hello
1 A abjdhfa sdf
2 b iuadbfa d
c a orbnbf
d a oiuertba sdf
e b adsofia eerygbd
f a aioihbkjs
g c oaihdufh
h a doihjksbsdfa
i b aoriufjkhadjfgo
j a dfk ahsddg a k n a fg
m a aioijdp poj n
n a opijsdkfk
poojith@Linux:~/Desktop/Techie$
```



```
poojith@Linux:~/Desktop/Techie$ awk '$2 == "a"' Hello
c a orbnbf
d a oiuertbasdf
f a aioihbkjs
h a doihjksbsdfa
j a dfkahsddgaknafg
m a aioijdpjohn
n a opijsdkfk
poojith@Linux:~/Desktop/Techie$
```

```
poojith@Linux:~/Desktop/Techie$ grep -i a Hello
1 A abjdhfasdf
2 b iuadbfa
c a orbnbf
d a oiuertbasdf
e b adsofiaeerygbd
f a aioihbkjs
g c oaihdufh
h a doihjksbsdfa
i b aoriufjkhadjfgo
j a dfkahsddgaknafg
m a aioijdpjohn
n a opijsdkfk
poojith@Linux:~/Desktop/Techie$
```

32. Calculate and print the average, sum, or other statistics of a column.

Calculate sum of column 3

```
awk '{sum += $3} END {print "Sum =", sum}' new_file
```

Calculate average of column 3

```
awk '{sum += $3; count++} END {print "Average =", sum/count}' new_file
```

```
poojith@Linux:~/Desktop/Techie$ ls
Fruits Fruits_hi.txt Hello Hi Linux New.tar Poojith alphabet compressfile delimfile new_file script.txt
poojith@Linux:~/Desktop/Techie$ cat new_file
101 a 50
102 b 60
103 c 80
104 e 60
105 d 80
106 f 70
poojith@Linux:~/Desktop/Techie$ awk '{sum += $3} END {print "Sum =",sum}' new_file
Sum = 400
poojith@Linux:~/Desktop/Techie$ awk '{sum += $3;count++} END {print "AVG =",sum/count}' new_file
AVG = 66.6667
poojith@Linux:~/Desktop/Techie$
```

Find minimum value in column 3

```
awk 'NR==1 || $3<min {min=$3} END {print "Minimum =", min}' new_file
```

Find maximum value in column 3

```
awk 'NR==1 || $3>max {max=$3} END {print "Maximum =", max}' new_file
```

```
poojith@Linux:~/Desktop/Techie$ ls
Fruits Fruits_hi.txt Hello Hi Linux New.tar Poojith alphabet compressfile delimfile new_file script.txt
poojith@Linux:~/Desktop/Techie$ cat new_file
101 a 50
102 b 60
103 c 80
104 e 60
105 d 80
106 f 70
poojith@Linux:~/Desktop/Techie$ awk '{sum += $3} END {print "Sum =",sum}' new_file
Sum = 400
poojith@Linux:~/Desktop/Techie$ awk '{sum += $3;count++} END {print "AVG =",sum/count}' new_file
AVG = 66.6667
poojith@Linux:~/Desktop/Techie$ awk 'NR==1 {max=$3} NR>1 && $3>max {max=$3} END {print "Maximum =", max}' new_file
Maximum = 80
poojith@Linux:~/Desktop/Techie$
```

33. Perform string manipulation, such as extracting substrings or changing case.

➔ Extracting

```
poojith@Linux:~/Desktop/Techie$ nano script.txt
poojith@Linux:~/Desktop/Techie$ cat script.txt
Hello World
DevOps Training
Linux Commands
AWS EC2
poojith@Linux:~/Desktop/Techie$ echo "Hello World" | cut -d ' ' -f 1
Hello
```

➔ Extract a substring using awk

```
poojith@Linux:~/Desktop/Techie$ cat script.txt
Hello World
DevOps Training
Linux Commands
AWS EC2
poojith@Linux:~/Desktop/Techie$ echo "Hello World" | awk '{print $1}'
Hello
poojith@Linux:~/Desktop/Techie$ echo "Hello World" | awk '{print $2}'
World
poojith@Linux:~/Desktop/Techie$
```

4

➔ Extract characters from a string

```
poojith@Linux:~/Desktop/Techie$ str="Devops"
poojith@Linux:~/Desktop/Techie$ echo "${str:0:3}"
Dev
poojith@Linux:~/Desktop/Techie$
```

➔ Change case using tr

```
poojith@Linux:~/Desktop/Techie$ str="Devops"
poojith@Linux:~/Desktop/Techie$ echo "${str:0:3}"
Dev
poojith@Linux:~/Desktop/Techie$ echo "hello world" | tr 'a-z' 'A-Z'
HELLO WORLD
poojith@Linux:~/Desktop/Techie$ echo "HELLO WORLD" | tr 'A-Z' 'a-z'
hello world
poojith@Linux:~/Desktop/Techie$
```

34. Count the occurrences of a specific pattern in a file.

grep Searches for a pattern

-o Prints **only the matching part**, not the entire line

"Linux" → Pattern we want to find

Linux → File to search

```
poojith@Linux:~/Desktop/Techie$ touch Linux
poojith@Linux:~/Desktop/Techie$ vi Linux
poojith@Linux:~/Desktop/Techie$ vi Linux
poojith@Linux:~/Desktop/Techie$ grep -o "Linux" Linux
Linux
Linux
Linux
Linux
Linux
poojith@Linux:~/Desktop/Techie$ grep -o "Linux" Linux | wc -1
error: unexpected argument '-1' found

tip: to pass '-1' as a value, use '-- -1'

Usage: wc [OPTION]... [FILE]...

For more information, try '--help'.
poojith@Linux:~/Desktop/Techie$ grep -o "Linux" Linux | wc -l
5
poojith@Linux:~/Desktop/Techie$
```

35. Sort lines based on a specific field or column.

Sort lines alphabetically sort file

```
poojith@Linux: ~/Desktop/Techie — vi alphabet
~/Desktop/Techie
a
b
c
d
e
f
g
h
i
j
k
l
m
n
o
p
q
r
s
t
u
v
w
x
y
z
"alphabet" 27 lines, 53 bytes
```

```

poojith@Linux:~/Desktop/Techie$ ls
Hello Hi Linux New.tar compressfile
poojith@Linux:~/Desktop/Techie$ touch alphabet
poojith@Linux:~/Desktop/Techie$ vi alphabet
poojith@Linux:~/Desktop/Techie$ vi alphabet
poojith@Linux:~/Desktop/Techie$ sort alphabet

a
b
c
d
e
f
g
h
i
j
k
l
m
n
o
p
q
r
s
t
u
v
w
x
y
z
poojith@Linux:~/Desktop/Techie$

```

Sort based on 1st column `sort -k1 file.txt`

```

poojith@Linux: ~/Desktop/Techie — vi Linux
~/Desktop/Techie
Linux is an open platform
Linux is easy to learn
i am leraning linux
Linux Linux Linux
~
~
~

poojith@Linux:~/Desktop/Techie$ sort -k1 Linux
i am leraning linux
Linux is an open platform
Linux is easy to learn
Linux Linux Linux
poojith@Linux:~/Desktop/Techie$ vi Linux
poojith@Linux:~/Desktop/Techie$

```

36. Merge multiple files based on a common field or column.

Purpose	Command
Merge two files using first column	join file1 file2
Sort files before joining	sort -k1,1 file1
Use column 2 from first file	join -1 2 file1 file2
Use column 2 from both files	join -1 2 -2 2 file1 file2

```
poojith@Linux:~/Desktop/Techie$ ls
Fruits Fruits_hi.txt Hello Hi Linux New.tar alphabet compressfile delimfile joinedfile.txt new_file
poojith@Linux:~/Desktop/Techie$ sort Fruits
1 Apple
2 Banana
3 Carrot
4 Mango
5 Grapes
6 Orange
7 Watermelon
poojith@Linux:~/Desktop/Techie$ sort Hi
Apple 50
Banana 40
Carrot 60
Grapes 80
Mango 50
Orange 40
Watermelon 65
```

```
poojith@Linux:~/Desktop/Techie$ sort -k1,1 Hi
```

```
Apple 50
Banana 40
Carrot 60
Grapes 80
Mango 50
Orange 40
Watermelon 65
```

```
poojith@Linux:~/Desktop/Techie$ sort -k2,2 Fruits
```

```
1 Apple
2 Banana
3 Carrot
5 Grapes
4 Mango
6 Orange
7 Watermelon
```

```
poojith@Linux:~/Desktop/Techie$ join -1 2 Fruits Hi > Fruits_hi.txt
```

```
poojith@Linux:~/Desktop/Techie$ cat Fruits_hi.txt
```

```
Apple 1 50
Banana 2 40
Carrot 3 60
Mango 4 50
Grapes 5 80
Orange 6 40
Watermelon 7 65
```

```
poojith@Linux:~/Desktop/Techie$
```

```
poojith@Linux:~/Desktop/Techie$ join -1 2 -2 1 Fruits Hi > Fruits_hi.txt
```

```
poojith@Linux:~/Desktop/Techie$ cat Fruits_hi.txt
```

```
Apple 1 50
Banana 2 40
Carrot 3 60
Mango 4 50
Grapes 5 80
Orange 6 40
Watermelon 7 65
```

37. Substitute text in a file using search and replace.

Replace first occurrence in each line `sed 's/old/new/' file.txt`

Replace all occurrences in each line `sed 's/old/new/g' file.txt`

Replace text and save changes to the file `sed -i 's/old/new/g' file.txt`

Replace only on a specific line `sed '5s/old/new/' file.txt`

Replace a word throughout the file `sed -i 's/Linux/Ubuntu/g' file.txt`

```
poojith@Linux:~/Desktop/Techie$ ls
Fruits Fruits_hi.txt Hello Hi Linux New.tar Poojith alphabet compressfile delinfile new_file
poojith@Linux:~/Desktop/Techie$ cat Poojith
Hi this is Poojith Ravutla,
Poojith is my name
iam from Nizamabad
Nizamabad is the City
my village is in Nizamabad District
Poojith
from- Nizamabad
Village- Mailaram
Mandal- Sirikonda
Dist- Nizamabad
poojith@Linux:~/Desktop/Techie$ sed 's/Poojith/Shodhan' Poojith
sed: -e expression #1, char 17: unterminated 's' command
poojith@Linux:~/Desktop/Techie$ sed 's/Poojith/Shodhan/' Poojith
Hi this is Shodhan Ravutla,
Shodhan is my name
iam from Nizamabad
Nizamabad is the City
my village is in Nizamabad District
Shodhan
from- Nizamabad
Village- Mailaram
Mandal- Sirikonda
Dist- Nizamabad
poojith@Linux:~/Desktop/Techie$
```

```
poojith@Linux:~/Desktop/Techie$ sed 's/Poojith/Shodhan/g' Poojith
Hi this is Shodhan Ravutla,
Shodhan is my name
iam from Nizamabad
Nizamabad is the City
my village is in Nizamabad District
Shodhan
from- Nizamabad
Village- Mailaram
Mandal- Sirikonda
Dist- Nizamabad
```

```
poojith@Linux:~/Desktop/Techie$ cat Poojith
Hi this is Poojith Ravutla,
Poojith is my name
iam from Nizamabad
Nizamabad is the City
my village is in Nizamabad District
Poojith
from- Nizamabad
Village- Mailaram
Mandal- Sirikonda
Dist- Nizamabad
poojith@Linux:~/Desktop/Techie$ sed -i 's/Poojith/Shodhan/g' Poojith
poojith@Linux:~/Desktop/Techie$ cat Poojith
Hi this is Shodhan Ravutla,
Shodhan is my name
iam from Nizamabad
Nizamabad is the City
my village is in Nizamabad District
Shodhan
from- Nizamabad
Village- Mailaram
Mandal- Sirikonda
Dist- Nizamabad
poojith@Linux:~/Desktop/Techie$
```



```
poojith@Linux:~/Desktop/Techie$ sed '4s/Nizamabad/Hyderabad/g' Poojith
Hi this is Shodhan Ravutla,
Shodhan is my name
iam from Nizamabad
Hyderabad is the City
my village is in Nizamabad District
Shodhan
from- Nizamabad
Village- Mailaram
Mandal- Sirikonda
Dist- Nizamabad
poojith@Linux:~/Desktop/Techie$
```

38. Delete specific lines based on a pattern or line number.

Purpose	Command
Delete a specific line	sed '5d' file.txt
Delete multiple specific lines	sed '2,5d' file.txt
Delete the first line	sed '1d' file.txt
Delete the last line	sed '\$d' file.txt
Delete lines containing a pattern	sed '/pattern/d' file.txt
Delete lines starting with a pattern	sed '/^pattern/d' file.txt
Delete lines ending with a pattern	sed '/pattern\$/d' file.txt
Delete lines between two patterns	sed '/start/,end/d' file.txt
Delete lines from the actual file	sed -i '5d' file.txt

```
poojith@Linux:~/Desktop/Techie$ cat Fruits
Apple
Banana
Carrot
Dragon
Orange
Mango
poojith@Linux:~/Desktop/Techie$ sed '5d' Fruits
Apple
Banana
Carrot
Dragon
Mango
poojith@Linux:~/Desktop/Techie$ sed '2,5d' Fruits
Apple
Mango
poojith@Linux:~/Desktop/Techie$ sed '1d' Fruits
Banana
Carrot
Dragon
Orange
Mango
poojith@Linux:~/Desktop/Techie$ sed '$d' Fruits
Apple
Banana
Carrot
Dragon
Orange
poojith@Linux:~/Desktop/Techie$
```

```

poojith@Linux:~/Desktop/Techie$ ls
Fruits Fruits_hi.txt Hello Hi Linux New.tar Poojith alphabet compressfile delimfile new_file
poojith@Linux:~/Desktop/Techie$ cat Fruits
1 Apple
2 Banana
3 Carrot
4 Mango
5 Grapes
6 Orange
7 Watermelon
poojith@Linux:~/Desktop/Techie$ sed -i '7d' Fruits
poojith@Linux:~/Desktop/Techie$ cat Fruits
1 Apple
2 Banana
3 Carrot
4 Mango
5 Grapes
6 Orange
poojith@Linux:~/Desktop/Techie$

```

39. Append or insert text before or after a specific pattern or line.

Insert text **before** a matching pattern `sed '/pattern/i\Text to insert' file.txt`

Insert text **after** a matching pattern `sed '/pattern/a\Text to insert' file.txt`

Insert text **before** a specific line number `sed '5i\Text to insert' file.txt`

Insert text **after** a specific line number `sed '5a\Text to insert' file.txt`

Modify the file directly `sed -i '/pattern/a\Text to insert' file.txt`

```

poojith@Linux:~/Desktop/Techie$ ls
0 Fruits Hello Hi Linux New.tar alphabet compressfile delimfile
poojith@Linux:~/Desktop/Techie$ sed '/Carrot/i\Mango' Fruits
Apple
Banana
/Mango
Carrot
Dragon
poojith@Linux:~/Desktop/Techie$ sed '3d' Fruits
Apple
Banana
Dragon
poojith@Linux:~/Desktop/Techie$ sed '/Carrot/i\Mango' Fruits
Apple
Banana
Mango
Carrot
Dragon
poojith@Linux:~/Desktop/Techie$ sed '2,5d' Fruits
Apple
poojith@Linux:~/Desktop/Techie$ cat Fruits
Apple
Banana
Carrot
Dragon
poojith@Linux:~/Desktop/Techie$ sed 'Dragon/a\Orange' Fruits
sed: -e expression #1, char 2: extra characters after command
poojith@Linux:~/Desktop/Techie$ sed '/Dragon/a\Orange' Fruits
Apple
Banana
Carrot
Dragon
Orange
poojith@Linux:~/Desktop/Techie$

```

40. Print only specific lines from a file.

- ➔ We use AWK to print only specific lines from a file
- ➔ For ex. We use `ps -ef`, it will give all the running processes details
- ➔ Now we use **awk** command to get specific lines from the output.
- ➔ We use `ps -ef | awk '{print $2}'`

```
poojith@Linux:~/Desktop/Techie/compressfile$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1        0  0  09:38 ?        00:00:13 /usr/lib/systemd/systemd --switched-root --system --deserialize=51
root           2        0  0  09:38 ?        00:00:00 [kthreadd]
root           3        2  0  09:38 ?        00:00:00 [pool_workqueue_release]
root           4        2  0  09:38 ?        00:00:00 [kworker/R-rcu_gp]
root           5        2  0  09:38 ?        00:00:00 [kworker/R-sync_wq]
root           6        2  0  09:38 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7        2  0  09:38 ?        00:00:00 [kworker/R-slub_flushwq]
root           8        2  0  09:38 ?        00:00:00 [kworker/R-netns]
root          12        2  0  09:38 ?        00:00:00 [kworker/u32:0-grtr_ns_handler]
root          13        2  0  09:38 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14        2  0  09:38 ?        00:00:00 [ksoftirqd/0]
root          15        2  0  09:38 ?        00:00:14 [rcu_preempt]
root          16        2  0  09:38 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17        2  0  09:38 ?        00:00:03 [rcu_exp_gp_kthread_worker]
root          18        2  0  09:38 ?        00:00:01 [migration/0]
root          19        2  0  09:38 ?        00:00:00 [kprobe-optimizer]
root          20        2  0  09:38 ?        00:00:00 [idle_inject/0]
root          21        2  0  09:38 ?        00:00:00 [cpuhp/0]
root          22        2  0  09:38 ?        00:00:00 [cpuhp/1]
root          23        2  0  09:38 ?        00:00:00 [idle_inject/1]
root          24        2  0  09:38 ?        00:00:01 [migration/1]
root          25        2  0  09:38 ?        00:00:02 [ksoftirqd/1]
root          28        2  0  09:38 ?        00:00:00 [cpuhp/2]
root          29        2  0  09:38 ?        00:00:00 [idle_inject/2]
root          30        2  0  09:38 ?        00:00:00 [migration/2]
root          31        2  0  09:38 ?        00:00:00 [ksoftirqd/2]
root          34        2  0  09:38 ?        00:00:00 [cpuhp/3]
root          35        2  0  09:38 ?        00:00:00 [idle_inject/3]
root          36        2  0  09:38 ?        00:00:00 [migration/3]
root          37        2  0  09:38 ?        00:00:00 [ksoftirqd/3]
root          39        2  0  09:38 ?        00:00:00 [kworker/3:0H-kblockd]
root          40        2  0  09:38 ?        00:00:00 [cpuhp/4]
root          41        2  0  09:38 ?        00:00:00 [idle_inject/4]
root          42        2  0  09:38 ?        00:00:00 [migration/4]
root          43        2  0  09:38 ?        00:00:00 [ksoftirqd/4]
root          46        2  0  09:38 ?        00:00:00 [cpuhp/5]
root          47        2  0  09:38 ?        00:00:00 [idle_inject/5]
root          48        2  0  09:38 ?        00:00:00 [migration/5]
root          49        2  0  09:38 ?        00:00:00 [ksoftirqd/5]
root          52        2  0  09:38 ?        00:00:00 [cpuhp/6]
root          53        2  0  09:38 ?        00:00:00 [idle_inject/6]
root          54        2  0  09:38 ?        00:00:00 [migration/6]
root          55        2  0  09:38 ?        00:00:00 [ksoftirqd/6]
root          58        2  0  09:38 ?        00:00:00 [cpuhp/7]
root          59        2  0  09:38 ?        00:00:00 [idle_inject/7]
root          8023        2  0  11:36 ?        00:00:00 [kworker/u45:1]
root          8024        2  0  11:57 ?        00:00:00 [kworker/u35:0-flush-8:0]
poojith       8855    3927  0  11:58 ?        00:00:00 ojs /usr/share/gnome-shell/extensions/dingrastersoft.com/app/ding.js -E -P /usr/share/gnome-shell/extensions/dingrastersoft.com/app
root          8976        2  0  11:58 ?        00:00:00 [kworker/3:2-cgroup_free]
root          8985        2  0  12:00 ?        00:00:00 [kworker/1:1-cgroup_free]
root          9062        2  0  12:07 ?        00:00:00 [kworker/u35:2-events_unbound]
root          9075        2  0  12:10 ?        00:00:00 [kworker/5:0-events]
root          9181        2  0  12:11 ?        00:00:00 [kworker/u34:0]
poojith       9160    5377  0  12:19 pts/0    00:00:00 ps -ef

poojith@Linux:~/Desktop/Techie/compressfile$ ps -ef | awk '{print $2}'
PID
1
2
3
4
5
6
7
8
12
13
14
15
16
17
18
19
20
21
22
23
24
25
28
29
30
31
34
35
36
37
8023
8024
8855
8976
8985
9062
9075
9181
9160
```

```
poojith@Linux:~/Desktop/Techie/compressfile$ ps -ef | awk '{print $2, $3, $4}'
PID PPID C
1 0 0
2 0 0
3 2 0
4 2 0
5 2 0
6 2 0
7 2 0
8 2 0
12 2 0
13 2 0
14 2 0
15 2 0
16 2 0
17 2 0
18 2 0
19 2 0
20 2 0
21 2 0
22 2 0
23 2 0
24 2 0
25 2 0
28 2 0
29 2 0
30 2 0
31 2 0
34 2 0
35 2 0
36 2 0
37 2 0
39 2 0
40 2 0
41 2 0
42 2 0
43 2 0
46 2 0
47 2 0
48 2 0
49 2 0
52 2 0
53 2 0
54 2 0
55 2 0
```

41. Copy file from linux to windows machine

- ➔ Connect to the AWS EC2 Linux instance
- ➔ Create a test file (test.txt)
- ➔ Open Git Bash on Windows – cd ~/Downloads
- ➔ Copy the file using SCP
- ➔ \$scp -i "mytest_key.pem" ec2-user@54.173.8.41:/home/ec2-user/test.txt ~/Downloads/
- ➔ Verify the file transfer (test.txt 100% 42 0.1KB/s 00:00)
- ➔ Verify the file on Windows – (ls -l | grep test.txt)

```
~/m/'
Last login: Tue Aug 11 06:51:28 2026 from 103.203.172.230
[ec2-user@ip-172-31-19-159 ~]$ ls -l /home/ubuntu/test.txt
ls: cannot access '/home/ubuntu/test.txt': No such file or directory
[ec2-user@ip-172-31-19-159 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-19-159 ~]$ ls -l
total 16
drwxr-xr-x. 9 ec2-user ec2-user 16384 Aug 11 11:01 jenkins
[ec2-user@ip-172-31-19-159 ~]$ find /home/ec2-user -name "test.txt"
[ec2-user@ip-172-31-19-159 ~]$ echo "This file is copied from Linux to Windows" > test.txt
[ec2-user@ip-172-31-19-159 ~]$ ls -l test.txt
-rw-r--r--. 1 ec2-user ec2-user 42 Aug 11 11:28 test.txt
[ec2-user@ip-172-31-19-159 ~]$
```

```
pooji@Poojith MINGW64 ~/Downloads
$ scp -i "mytest_key.pem" ec2-user@54.173.8.41:/home/ec2-user/test.txt ~/Downloads/
The authenticity of host '54.173.8.41 (54.173.8.41)' can't be established.
ec25519 key fingerprint is SHA256:CEv03qJ+123xxeN1ieeuokQ25frpMuhtYSo8zdT8eJ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:2: ec2-100-55-86-132.compute-1.amazonaws.com
  ~/.ssh/known_hosts:4: ec2-54-242-216-9.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.173.8.41' (ec25519) to the list of known hosts.
test.txt 100% 42 0.1KB/s 00:00
```

```
pooji@Poojith MINGW64 ~/Downloads
$ ls -l | grep test.txt
-rw-r--r-- 1 pooji 197609 42 Aug 11 17:00 test.txt

pooji@Poojith MINGW64 ~/Downloads
$
```

42. 5 use cases for AWK and 5 use cases for sed

Use case	Purpose	Command
Print specific columns	Extract required fields from a file	<code>awk '{print \$1, \$3}' file.txt</code>
Filter rows based on condition	Print only matching records	<code>awk '\$3 > 100 {print \$1, \$3}' file.txt</code>
Calculate sum/average	Perform calculations on columns	<code>awk '{sum += \$3} END {print sum}' file.txt</code>
Count occurrences	Count lines/records matching a pattern	<code>awk '/ERROR/ {count++} END {print count}' app.log</code>
Process delimited files	Work with CSV/colon-separated files	<code>awk -F',' '{print \$1, \$3}' users.csv</code>

```
poojith@Linux:~/Desktop/Techie$ ls
Hello Hi Linux New.tar alphabet compressfile delimfile
poojith@Linux:~/Desktop/Techie$ cat delimfile
100,Poojith,ABC
101,Ravutla,DEF
102,Mailaram,GHI
103,Sirikonda,JKL
104,Nizamabad,MNO
poojith@Linux:~/Desktop/Techie$ awk '{print $1, $2}' delimfile
100,Poojith,ABC
101,Ravutla,DEF
102,Mailaram,GHI
103,Sirikonda,JKL
104,Nizamabad,MNO
poojith@Linux:~/Desktop/Techie$ awk -f',' '{print $1, $2}' delimfile
awk: cannot open ""," (No such file or directory)
poojith@Linux:~/Desktop/Techie$ awk -F',' '{print $1, $2}' delimfile
100 Poojith
101 Ravutla
102 Mailaram
103 Sirikonda
104 Nizamabad
poojith@Linux:~/Desktop/Techie$ awk $>0 {print $1, $3}
awk: line 1: syntax error at or near end of line
poojith@Linux:~/Desktop/Techie$ awk $>0 {print $1, $3} delimfile
awk: line 1: syntax error at or near end of line
poojith@Linux:~/Desktop/Techie$ awk '$1>0 {print $1, $3}' delimfile
100,Poojith,ABC
101,Ravutla,DEF
102,Mailaram,GHI
103,Sirikonda,JKL
104,Nizamabad,MNO
poojith@Linux:~/Desktop/Techie$ awk -F',' '$1>0 {print $1, $3}' delimfile
100 ABC
101 DEF
102 GHI
103 JKL
104 MNO
poojith@Linux:~/Desktop/Techie$ awk '{sum += $3}END{print sum}' delimfile
0
poojith@Linux:~/Desktop/Techie$ awk '{sum += $1}END{print sum}' delimfile
510
poojith@Linux:~/Desktop/Techie$
```

```
poojith@Linux:~/Desktop/Techie$ awk '/Linux/{count++}END{print count}' Linux
3
poojith@Linux:~/Desktop/Techie$ cat Linux
Linux is an open platform
Linux is easy to learn
i am leraning linux
Linux Linux Linux
poojith@Linux:~/Desktop/Techie$
```